
Managerial Economics

Premium Hinglish Study E-Book

Complete Exam Guide — Concepts, Intuition, Derivations & PYQs

Units 1–4 (Weeks 1–4) — Complete Coverage

Unit 1 Foundations of Managerial Economics

Unit 2 Costs, Production & Profit Maximisation

Unit 3 MRTS, Cost Minimisation, Returns to Scale & Supply

Unit 4 Short-/Long-Run Costs & Consumer Theory

Ek teacher ki tarah har concept ko samjhaane wali notes — bilkul exam-ready, Hinglish mein.

IIT Madras — BS in Data Science & Applications

Course: Managerial Economics (Prof. Vimal Kumar)

Made by Anmol Kansal using AI

“Economics is a science of decision-making.”

About This E-Book

Yeh ek single, premium textbook-quality study guide hai jo IIT Madras ke *Managerial Economics* course ke **Weeks 1–4** ko cover karti hai. Iska maksad simple hai: har concept ko aise samjhaana jaise koi accha teacher class mein samjhaata hai — Hindi mein explanation, technical terms English mein.

Is poori book mein *kuch bhi summarise nahi kiya gaya*; saara original content, examples, tables, derivations, diagrams (TikZ se recreate), aur exam-tips waise ke waise preserve kiye gaye hain, plus har topic ke saath relevant **PYQ boxes** taaki exam connection clear rahe.

Kaise padhein: Pehle Unit sequence mein padho — Unit 1 buyer/decision-making, Unit 2 cost & production, Unit 3 cost-minimisation & supply, Unit 4 consumer theory. Har Unit ke end mein *Quick Revision*, *Expected Questions* aur *One-Night-Before* notes hain.

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Made by Anmol Kansal using AI — For IITM BS, DS & Applications.

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Unit 1

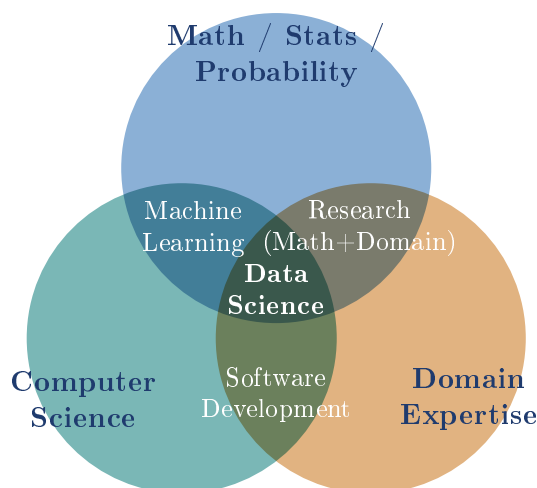
Foundations of Managerial Economics

1.1 Why Study Economics? — The Big Picture

Intuition

Tum **Managerial Economics** ek *Data Science* programme ke part ke roop mein padh rahe ho. Sawaal yeh hai — *kyun?* Kyunki ek accha data scientist banne ke liye *teen* cheezein ek saath chahiye: mathematics/statistics, computer science, *aur* domain expertise. Economics shayad data scientist ke liye sabse powerful domain hai, kyunki yeh khud hi quantitative bhi hai, social bhi, aur causal reasoning se bhari hui bhi.

Prof. Kumar course ki shuruaat ek **Venn diagram** se karte hain — teen circles: mathematics & statistics, computer science, aur domain expertise. Jahaan koi bhi do circles overlap karte hain, wahaan ek specialised field milti hai; aur jahaan teeno milte hain, wahaan banta hai **Data Science**.



1.1.1 What is Economics? — Multiple Lenses

Teen complementary definitions yaad rakhni hain (teeno ko ek saath internalise karo):

1. **Decision-making ka science.** Har pal hum alag-alag alternatives mein se choose karte hain; economics in choices ke peeche ke *logic* ko study karti hai.
2. **(Marshall, 1885)** “A study of human behaviour in the *ordinary business of life*.” — yahaan *ordinary* shabd crucial hai: hum regular, rozmarra ke decisions model karte hain, koi ajeeb-o-gareeb exceptions nahi.
3. **Scarce resources ko allocate karke individual wants/desires ko satisfy karna.** Chaar keywords hain: *allocating*, *scarce*, *individual*, *wants/desires*.

Core Definition — Economics

Economics woh framework hai jo study karta hai ki decision-making units (individuals, households, firms, governments) apne **scarce** resources ko **unlimited wants and desires** ke beech kaise allocate karte hain. Mathematically, yeh ek applied course hai **constrained optimisa-**

tion ka.

Why Scarcity Matters

Agar resources unlimited hote, toh koi choice karne ki zaroorat hi nahi padti — sabke paas sab kuch hota. Economics exist hi isliye karti hai kyunki *unlimited wants* aur *limited means* (paisa, time, raw materials, attention) ke beech ek gap hamesha rehta hai. Yahaan tak ki Mukesh Ambani ke paas bhi ek scarce resource hai: *time*. Isliye duniya ka sabse ameer aadmi bhi har din economic choices karta hai.

1.1.2 What is Managerial Economics?

Definition — Manager & Managerial Economics

Ek **Manager** woh vyakti hai jo resources ko direct karke ek defined goal achieve karta hai.

Managerial Economics woh study hai ki scarce resources ko aise kaise direct karein ki managerial goal sabse *efficiently* achieve ho. Doosre shabdon mein, yeh economics (mukhyatah microeconomics) ka business problems par application hai.

Economics ki teen branches:

- **Microeconomics** — individual decision-makers ki study (ek household, ek firm, ek consumer).
- **Macroeconomics** — aggregate phenomena ki study (inflation, unemployment, GDP, interest rates).
- **Econometrics** — economic phenomena ka measurement, jismein statistics aur economic theory dono milte hain; khaaskar observational data se causal inference.

Managerial Economics lagbhag poori tarah **microeconomics** se draw karti hai.

Microeconomics ke teen players: Producers / Firms (sellers) — Consumers / Buyers — aur woh **Market** jahaan yeh dono milte hain (real ho ya virtual).

1.1.3 Why Econometrics is Special (Brief)

Classical statistics mein (jaise physics ke experiments) tum X ko control karte ho aur Y observe karte ho — independent vs. dependent variable ka strong notion. Economics mein tum “education ko vary” waise nahi kar sakte jaise pendulum ki length vary karte ho: yahaan X aur Y dono *stochastic* hote hain. Isiliye econometrics — jo observational data se causal inference par focus karti hai — textbook statistics se alag hai.

1.2 The Tools We Will Use

Remember This

Is poore course mein chaar mathematical/statistical tools chhaaye rehte hain:

- (1) **Constrained Optimisation**,
- (2) **Equilibrium Analysis**,
- (3) **Comparative Statics**,
- (4) **Regression Analysis** (halka-fulka).

1.2.1 Variables, Parameters, and Models

Ek **model** reality ka ek simplified representation hota hai. Jaise ek metro map buildings ko ignore karke sirf stations highlight karta hai, waise hi hamare economic models inessential details ko discard kar dete hain.

- **Exogenous variables** — model ke bahar se diye gaye (e.g. ek chhoti firm ke liye raw materials ki price).
- **Endogenous variables** — model ke andar determine hote hain (e.g. firm kitna raw material kharidne ka decide karti hai).
- **Parameters** — environment ke fixed features (depreciation rate, contract se set worker wages).

1.2.2 Tool 1 — Constrained Optimisation

Generic problem:

$$\max_x f(x) \quad \text{s.t.} \quad g(x) \leq c$$

yahaan $f(x)$ **objective function** hai (jise tum maximise karte ho — e.g. profit ya utility), x tumhara **choice variable** hai, aur $g(x) \leq c$ tumhara **constraint** hai (limited resources). Minimisation problems (e.g. cost minimisation) bilkul symmetric hote hain.

- Consumers **utility** (satisfaction ka level) maximise karte hain.
- Managers / firms **profit** maximise karte hain (ya cost minimise).

1.2.3 Tool 2 — Equilibrium

Equilibrium opposing forces ka balance hai: jaise buyers chaahte hain kam price, sellers chaahte hain zyada price. *Price* woh coordinating mechanism hai. Equilibrium mein kisi bhi agent ke paas apna behaviour badalne ka incentive nahi hota.

1.2.4 Tool 3 — Comparative Statics

Ek equilibrium ki “photograph” lo, ek exogenous variable change karo, phir naye equilibrium ki “photograph” lo, aur dono ko compare karo — yahi **comparative statics** hai. Tum beech ka transition ignore kar dete ho (woh comparative dynamics hota, ek macro topic).

Exam Tip

Comparative statics = do equilibrium photographs ko compare karna.
Comparative dynamics = un dono ke beech ki video study karna.
Yeh course sirf *statics* par focus karta hai.

1.2.5 Tool 4 — Regression Analysis

Demand estimation ke liye firm data collect karke ek regression run karti hai: $Y = X\beta + \varepsilon$. Ek deterministic equation ke ulat, yahaan error term ε unobserved factors ko capture karta hai. Hum regression ko sirf halke se touch karenge.

1.3 The Apartment-Rental Problem — Demand ko Scratch se Banana

Real-World Application

Story. Tumhare paas ek apartment complex hai jismein **5 khaali units** hain. **Das** potential tenants aate hain, har ek ka apna maximum willingness-to-pay (WTP, hazaar rupaye per month):

Person	Max WTP (Rs '000 / month)
1	70
2	22
3	55
4	15
5	22
6	35
7	55
8	50
9	60
10	65

Sawaal: Maximum (aur minimum) rent kya hoga jis par saare 5 apartments rent ho jaayein?

1.3.1 Brute-Force Reasoning

WTPs ko ghatte hue (decreasing) order mein sort karo:

70, 65, 60, 55, 55, 50, 35, 22, 22, 15.

Ek rent P har us tenant ko attract karta hai jiska $WTP \geq P$ hai.

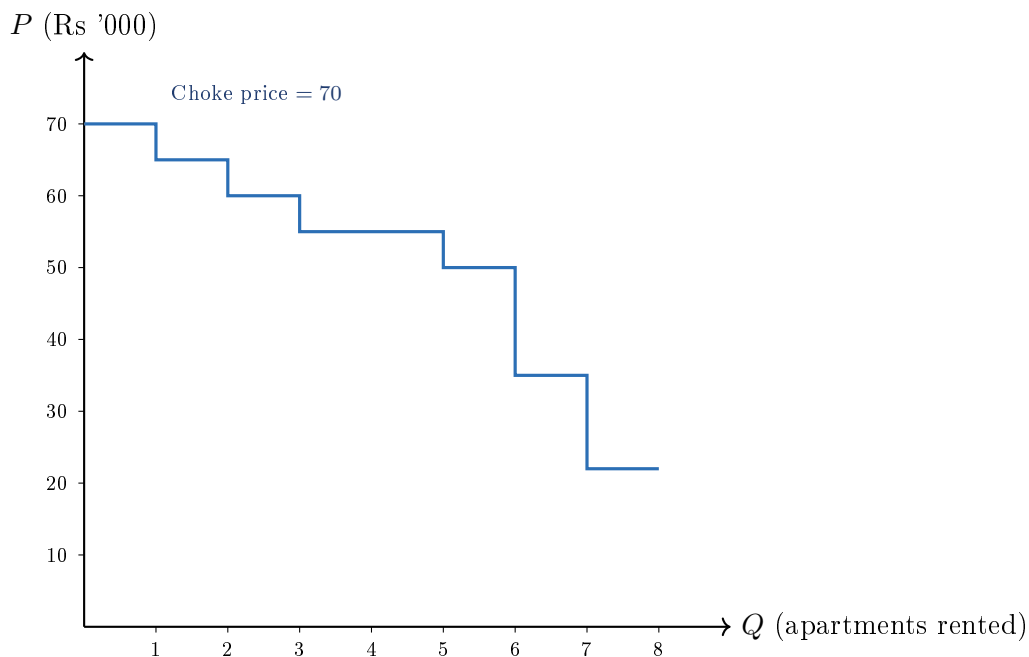
- $P > 70$: koi nahi.
- $65 < P \leq 70$: sirf person 1.
- $60 < P \leq 65$: persons 1, 10 (2 tenants).
- $55 < P \leq 60$: persons 1, 10, 9 (3 tenants).
- $50 < P \leq 55$: persons 1, 3, 7, 10 (4 tenants — do log 55 par tied hain).
- $35 < P \leq 50$: persons 1, 3, 7, 8, 10 (**5 tenants!**) ✓
- $22 < P \leq 35$: 6 tenants. Aur aage isi tarah.

Remember This

Answer. Exactly 5 apartments rent karne ke liye rent $P \in (35, 50]$ hazaar rupaye rakho. Maximum rent Rs 50,000 hai aur agla breakpoint neeche Rs 35,000 par hai.

1.3.2 Demand Function tak Generalise Karna

Har price P ke liye, count karo ki kitne logon ka $WTP \geq P$ hai. Woh count hi quantity demanded $Q(P)$ hai. P ko y-axis par (Marshall, 1885 se chali aa rahi tradition) aur Q ko x-axis par plot karne se ek **step-function demand curve** banti hai.



Definitions — Demand Vocabulary

- **Reservation price** (buyer ka): ek additional unit ke liye buyer jitna maximum dene ko taiyaar hai. (Yahaan “Max WTP” ke barabar.)
- **Choke price**: woh price jiske upar koi kuch nahi kharidta. (Yahaan: 70.)
- **Demand function** $Q(P)$: har price P ko quantity demanded se map karta hai.
- **Inverse demand function** $P(Q)$: har quantity Q ko us price se map karta hai jis par woh quantity clear hoti hai.

1.3.3 Step Function ko Straight Line se Approximate Karna

Agar WTPs dense ho jaayein, toh steps chhote ho jaate hain aur curve smooth ho jaati hai. Ek common smooth approximation:

$$Q = \max\{30 - P, 0\}, \quad \text{ya equivalently} \quad P = 30 - Q \quad (0 \leq Q \leq 30).$$

Formula Insight

$\max\{\cdot, 0\}$ wrapper bahut zaroori hai: iske bina, $P = 40$ daalne par $Q = 30 - P = -10$ aata hai, jo economically meaningless hai. Quantity kabhi negative nahi ho sakti.

Common Mistake

Bahut se students **demand** aur **quantity demanded** ko confuse kar dete hain.

- Sirf price change hone se *usi demand curve par movement* hoti hai — yeh **quantity demanded mein change** hai.
- Kisi aur factor (income, taste, substitutes/complements ki price) ke change se *poori curve shift* hoti hai — yeh **demand mein change** hai.

1.4 Modelling the Buyer — Teen Behavioural Assumptions

Hum har buyer ko ek **rational decision-maker** maante hain. Yahaan rationality ka matlab hai *choices ki consistency*, perfection nahi.

1. Har buyer ke paas har unit ke liye ek (subjective) reservation price r hai.
2. Buyer price P par kharidte waqt **consumer surplus** $r - P$ maximise karta hai.
3. Alternatives mein se buyer woh option chunta hai jismein sabse zyada surplus ho.

Consumer Surplus

Ek buyer jiski reservation price r hai aur jo market price P par kharidta hai:

$$\text{Consumer Surplus (CS)} = r - P.$$

Yeh transaction se hone wala gain hai (money-equivalent units mein). Buyer tabhi participate karta hai jab $CS \geq 0$ ho, aur woh sabse zyada CS wala alternative chunta hai.

1.4.1 Why “Rational”? — Ek Defence

Rationality ki assumption humein behaviour predict karne deti hai. Agar koi observed action irrational lage, toh hum aam taur par framework chhodne ke bajaaye koi *hidden context* dhoondte hain. Zyadatar decisions itne consistent hote hain ki yeh model useful rehta hai.

1.4.2 Reservation Price Consumption ke Saath Ghatti Hai

Agar tumne is summer abhi tak ek bhi mango nahi khaaya, toh pehle aam ke liye tumhari reservation price shayad Rs 100 ho. Us din ke doosre aam ke liye shayad sirf Rs 60. Reservation price ek additional unit ka maximum WTP hai, aur additional value aam taur par girti hai — yahi **Law of Diminishing Marginal Utility** (Week 4) ka beej hai.

1.5 Limitations of the Rationality Assumption

Common Mistake — Teen Classic Violations

Teen classic violations jo students ko pata honi chahiye:

1. **Context-dependent reservation prices.** Wahi cold drink: tum ek 5-star hotel se mangwaane par zyada doge, ek bori (sack) se kam. Good identical hai; jo price tum accept karte ho woh nahi.
2. **Percentage vs. absolute thinking.** Log Rs 8000 ke Kindle par Rs 1000 (12.5%) bachaane ke liye 1 km drive kar lete hain, par Rs 80,000 ke iPad par (1.25%) nahi — jabki absolute saving same hai.
3. **Reference prices instead of true CS maximisation.** Buyers aksar saari options compare karne ke bajaaye ek yaad kiya hua “fair price” use kar lete hain.

Yeh rationality ke *descriptive failures* hain aur **behavioural economics** ki neev hain — syllabus se bahar, par jaanna useful hai.

1.6 Modelling the Seller — The Profit Motive

1.6.1 Profit, Revenue, Cost

Formula Insight — Profit Identity

$$\pi = TR - TC, \quad TR = P \cdot Q, \quad TC = C(Q).$$

- π (Greek pi) = profit.
- TR = total revenue = price per unit $\times$ quantity sold.
- TC = total cost, Q ka function (zyada output $\Rightarrow$ aam taur par zyada cost).

Profit badhaane ke do tareeke. Seller ya toh (a) price badhaye (zyada margin) par aam taur par kam beche, ya (b) price ghataaye (zyada volume) par per unit kam kamaye. Sahi mix market environment par depend karta hai.

1.6.2 Per-Unit Price ko Affect Karne Wali Cheezein

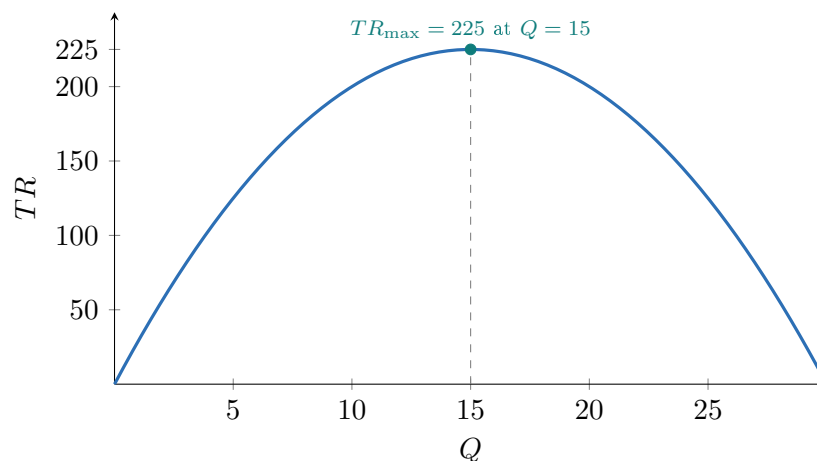
- **Discount** — early payment ke liye pehle hi diya jaata hai (e.g. “1/10, net 30” = 10 din mein pay karo toh 1% off; pura amount 30 din mein due).
- **Rebate** — purchase ke baad diya jaata hai agar buyer paperwork complete kare. Psychological idea: bahut buyers kabhi claim hi nahi karte.
- **Returns** — newspaper vendors unsold copies partial/full credit ke liye return kar sakte hain.
- **Allowance** — supermarket: “Mera product prominently display karo aur main price discount kar dunga.”

1.7 The Revenue Function and the Inverted-U

Linear inverse demand $P = 30 - Q$ ke saath, total revenue:

$$TR(Q) = P \cdot Q = (30 - Q)Q = 30Q - Q^2.$$

Yeh ek downward-opening parabola hai. $TR'(Q) = 30 - 2Q = 0$ set karne se revenue-maximising quantity $Q^* = 15$ milti hai, aur maximum revenue $TR_{\max} = 15 \cdot 15 = 225$.



Intuition

Inverted-U shape ek fundamental pattern hai: bahut kam Q par revenue chhota hai kyunki kam units bikte hain; bahut zyada Q par price itni neeche gir chuki hoti hai ki revenue collapse kar jaata hai. Maximum revenue kahin beech mein hota hai.

1.8 Market Structures — A Preview

Structure	Setting	Pricing power
Perfect competition	Bahut sellers, bahut buyers, identical product	None — har firm <i>price taker</i> ; price = marginal cost.
Monopoly	Ek seller, bahut buyers	Maximum — phir bhi demand curve se constrained.
Monopsony	Bahut sellers, ek buyer	Maximum — buyer ki taraf.
Oligopoly	Kuch sellers, bahut buyers	Strategic: har seller doosron ko anticipate karta hai (Game Theory!).

Remember This

Perfect competition mein firms ek doosre ko tab tak undercut karte rehte hain jab tak $P = MC$ na ho jaaye. MC se neeche, agli unit paisa lose karti hai, isliye koi aur price drop nahi karta.

1.9 Pricing Techniques — The Big Picture

Har transaction ek economic value create karta hai jo barabar hai (reservation price) — (marginal cost). Pie ka size in dono se fix hota hai; pie ka *division* bargaining power par depend karta hai.

Common techniques:

1. **Negotiation (haggling):** buyer aur seller mol-bhaav karte hain.
2. **Take-it-or-leave-it (fixed price):** seller pehle se ek price par commit kar deta hai.
3. **Auction:** seller buyers ki valuations reveal karwata hai.
4. **Price discrimination:** same product, alag buyers ko alag prices (hard-cover vs. paperback; advance vs. same-day airline tickets).

1.10 Sonu Nigam vs. Shaan Concert — Opportunity Cost in Action**Real-World Application**

Story. Kal tumhe Sonu Nigam ke concert ka ek *free* ticket mila (market price Rs 7500, resale value Rs 5000). Aaj pata chalta hai ki Shaan usi shaam ko concert de raha hai; uska ticket bhi Rs 7500 hai. Agar tumhe kal pata hota, toh tum Shaan ka ticket khareedte aur Sonu skip karte.

Sawaal. Kya tumhe Shaan ka concert attend karna chahiye aur Sonu ka nahi?

1.10.1 Dono Choices ko Surplus Comparison ki Tarah Frame Karna

Maan lo B_{SN} aur B_{SG} tumhara perceived benefit (rupaye mein) hai Sonu Nigam aur Shaan dekhne ka, respectively. Original information batati hai:

$$B_{SG} - 7500 > B_{SN} - 7500 \iff B_{SG} > B_{SN}.$$

Achha — par abhi conclusive nahi. Free Sonu ticket ab cost structure badal deta hai:

- Sonu Nigam jaane ka net benefit: $B_{SN} - 0 = B_{SN}$ (ticket already free).
- Shaan jaane ka net benefit: $B_{SG} - (7500 - 5000) = B_{SG} - 2500$, kyunki tum Sonu ticket Rs 5000 mein resell karke woh Shaan ke ticket mein laga sakte ho.

Tumhe Shaan tabhi attend karna chahiye jab $B_{SG} - 2500 > B_{SN}$, yaani $B_{SG} - B_{SN} > 2500$.

Remember This

Answer (shabdon mein). Yeh zaroori nahi hai. Tumhe Shaan tabhi attend karna chahiye jab Shaan ka Sonu ke upar perceived extra benefit Rs 2500 se zyada ho — kyunki Rs 2500 Sonu ticket ki resale value chhodne ki *opportunity cost* hai.

The Bigger Lesson — Opportunity Cost

Kisi action ki **opportunity cost** woh value hai jo next-best foregone option ki hai. Shaan jaana tumhe woh Rs 5000 “cost” karta hai jo tum Sonu ticket resell karke pa sakte the. Yeh idea hum Week 2 mein *explicit vs. implicit cost* ke roop mein formalise karenge.

1.11 Generalised Decision-Making Framework

Kisi bhi decision-maker (buyer ya seller) ke liye, maan lo x_i alternative i hai jiska benefit $b(x_i)$ aur cost $c(x_i)$ hai. Choose karo:

$$x^* = \arg \max_i [b(x_i) - c(x_i)].$$

- **Buyer** ke liye, b perceived value hai (money-equivalent terms mein) aur c price plus koi bhi opportunity cost.
- **Seller** ke liye, b total revenue hai aur c total cost.

Exam Tip — Happiness ko Money mein kyun badalte hain

Ek buyer ki “happiness” subjective hai, par hum pooch sakte hain: X ko avoid karne (ya enjoy karne) ke liye maximum kitna paisa chhodoge? Woh amount us experience ka *money-equivalent* hai. (Hypothetical: kya tum ek mahine jail se bachne ke liye Rs 5000 doge? Rs 1 lakh? Koi na koi price tumhari true disutility reveal kar deti hai.)

1.12 How to Write This Unit in the Exam

Answer-Writing Strategy

- **Important keywords:** scarcity, allocation, constrained optimisation, reservation price, consumer surplus ($r - P$), choke price, opportunity cost, sunk cost, equilibrium, comparative statics, price-taker.
- **Diagram suggestions:** (i) step-function demand curve + choke price, (ii) inverted-U total revenue parabola with $Q^* = a/2b$ marked, (iii) market-structure table. Har diagram

par axes label karna na bhoolo — P y-axis par, Q x-axis par.

- **Definition pehle, example baad mein:** pehle crisp formal definition likho, phir ek choti real-life example se support karo (mango wali, apartment wali).

Common Mistakes to Avoid

- “Demand” aur “quantity demanded” ko mat mila do — shift vs. movement.
- Free/non-refundable ticket wale sawaal mein *sunk cost* ko cost mat samjho; asli cost *resale (opportunity) value* hoti hai.
- Revenue maximise karte waqt $TR'(Q) = 0$ lagana mat bhoolo; sirf TR ki value likh dena galat hai.

1.13 Unit Summary — Quick Revision

Week 1 Quick-Revision Sheet

Big Ideas:

- Economics = decision-making ka science = constrained optimisation.
- Microeconomics producers, consumers, aur unke markets ko study karti hai.
- Chaar mathematical tools: optimisation, equilibrium, comparative statics, regression.

Buyer Side:

- Reservation price r = ek additional unit ka max WTP.
- Consumer surplus $CS = r - P$. Buyer sabse zyada CS wala option chunta hai.
- Demand curve downward-sloping; choke price wahaan jahaan woh y-axis ko hit karti hai.

Seller Side:

- Profit $\pi = TR - TC$, $TR = P \cdot Q$.
- Linear demand $P = a - bQ$ par TR ek inverted parabola hai.
- Revenue tab maximise hoti hai jab $TR'(Q) = 0$; $TR = aQ - bQ^2$ ke liye $Q^* = a/(2b)$.

Market Structures: Perfect competition (price-taker), Monopoly, Monopsony, Oligopoly.

Key Insight: Opportunity cost = next-best forgone option ki value.

Most Important Exam Topics

1. Managerial economics, opportunity cost, reservation price, consumer surplus, choke price ki definitions.
2. Demand curve par movement vs. shift.
3. r aur P diye hone par consumer surplus compute karna.
4. Rent ranges identify karna (apartment-style table) jahaan n buyers participate karein.
5. Decide karna ki sunk-cost (free ticket) involve hone par event attend karna hai ya nahi.
6. Linear demand curve ke liye TR maximise karne wala Q^* nikalna.

Expected Long Questions

1. “Economics” ki teeno definitions explain karo aur batao scarcity kyun zaroori hai; ek example ke saath constrained optimisation se link karo.
2. Apartment-rental example se ek demand curve construct karo aur choke price, reservation price, demand vs. quantity demanded explain karo.
3. Sonu vs. Shaan problem ko opportunity-cost framework se solve karo, har choice ko surplus comparison ki tarah frame karte hue.

Expected Short Questions

1. Reservation price aur choke price define karo.
2. Consumer surplus ka formula likho aur ek numerical example do.
3. Chaar mathematical tools list karo jo course mein use hote hain.
4. Perfect competition aur monopoly ka ek-ek difference batao.

One-Night-Before-Exam Revision Notes

- $CS = r - P$; $\pi = TR - TC$; $TR = P \cdot Q$.
- Linear demand $P = a - bQ \Rightarrow TR_{\max}$ at $Q^* = a/(2b)$.
- Apartment table: WTP ko sort karo, n -th highest se ek niche ka breakpoint dhoondo.
- Sunk cost ko ignore; resale/opportunity value hi asli cost.
- Comparative statics = do photos compare; statics only is course mein.

Practice Questions — Week 1

1. 6 buyers ke WTPs $\{80, 70, 65, 50, 30, 20\}$ aur 3 vacant apartments diye hain; woh rent range nikaalo jis par teeno rent ho jaayein.
2. Inverse demand $P = 40 - 2Q$ ke liye, revenue-maximising Q^* aur $TR_{\max}$ compute karo.
3. Batao har case mein *quantity demanded* mein change hai ya *demand* mein: (a) petrol ki price badhi; (b) ek nayi study coffee ko fashionable bana deti hai; (c) consumers ki income double ho jaati hai.
4. Ek buyer ki reservation price Rs 800 hai aur market price Rs 950. Uska consumer surplus kya hai? Kya woh kharidegi?
5. Tumhare paas Rs 2000 ka non-refundable concert ticket hai. Kal raat, ek dost tumhe ek alag concert ke free tickets offer karta hai jo tumhare liye Rs 3000 worth ke hain. Tum kaunsa attend karoge? Kyun? (Hint: Rs 2000 sunk hai.)

Unit 2

Costs, Production, and Profit Maximisation

2.1 Motivating Problem — Delhi Drive Karein ya Train Lein?

Real-World Application

Story. Tum Kanpur mein rehte ho (≈ 500 km Delhi se) aur driving vs. train ke beech sirf *cost* ke alawa indifferent ho. Train ka ticket Rs 1000 ka hai. Ek rental agency tumhe tumhari car ka yearly cost breakdown deti hai, yeh maan kar ki saal mein 20,000 km chalti hai:

Item	Annual (Rs)	Per km (Rs)
Insurance	10,000	0.50
Interest payment	20,000	1.00
Fuel & oil	10,000	0.50
Maintenance	10,000	0.50
Total	50,000	2.50

Naive per-km total Rs 2.50 hai, toh 500-km trip ka cost $500 \times 2.50 = \text{Rs } 1250$ nikalta hai — jisse train (Rs 1000) sasti lagti hai.

Quiz Trap

Yeh naive answer **galat** hai! Galti yeh hai ki har cost ko “variable” maan liya, jabki bahut saare costs *fixed* ya *sunk* hain (insurance aur interest toh chalo ya na chalo, dene hi hain). Is chapter ko padho — is problem par hum Week 4 mein wapas aayenge.

2.2 Cost Concepts — Explicit, Implicit, Sunk

Key Cost Definitions

- **Explicit cost** — firm dwara kiya gaya seedha monetary payment (wages, rent, bijli ka bill, grocers ko diya gaya paisa).
- **Implicit cost (opportunity cost)** — firm ke apne owned resources ki value jo business mein lagaye gaye, aur jinki *next-best use* chhod di gayi (e.g. apna business chalaane ke liye jo salary tum chhodte ho; apni hi capital invest karne par jo interest tum kama sakte the).
- **Sunk cost** — woh cost jo already ho chuki hai aur recover nahi ho sakti, chahe aage kuch bhi karo.
- **Opportunity cost of an action** — us action ko karne mein chhodi gayi next-best alternative ki value. Ismein explicit aur implicit dono cost shaamil hote hain.

2.2.1 Accounting Profit vs. Economic Profit

	Accounting Profit	Economic Profit
Definition	$TR - \text{explicit costs}$	$TR - \text{explicit} - \text{implicit costs}$
Opportunity cost included?	No	Yes
Used by	Accountants, tax authorities	Economists, decision-makers
Decision relevance	Limited	Direct

Why “Economic Profit” is the Right Concept

Maan lo tum ek shop chalate ho jiska accounting profit Rs 8 lakh/year hai. Agar tum ek salaried employee ke roop mein Rs 10 lakh/year kama sakte the, toh tumhari implicit cost Rs 10 lakh hai, aur tumhara *economic profit* = $-\text{Rs } 2 \text{ lakh}$. Iska matlab shop band karke job karna better hota. Accounting profit yeh sach chhupa leta hai.

2.2.2 Cost Analysis ke Do Cardinal Lessons

Remember This

- (1) Har decision mein *implicit (opportunity) cost* consider karo.
- (2) Sunk costs ko *ignore* karo — woh future decisions ke liye irrelevant hain.

Example — The Fritos Problem (PYQ classic). Tumne \$100 mein 5 cases Fritos kharide jo customer ne reject kar diye. Grocer return par \$0.25 per dollar (yaani total \$25) refund dega. Ek fraternity poore lot ke liye \$55 offer karti hai.

- Original purchase ka sunk cost: \$100 (*irrelevant!*).
- Fraternity ko bechne ki opportunity cost = next-best foregone = \$25 (grocer ko return).
- Frat sale se net gain = $55 - 25 = +\$30 > 0$. **Bech do!**

PYQ Family — Explicit / Implicit / Sunk (Snack-Delivery, 2025T1 Q208–209, Jan 2026 Q22–)

Story. Tum campus par ek snack-delivery business start karte ho. Web-hosting ke \$500/month dete ho aur ek SUV own karte ho jiski \$300 monthly car payment aur \$100/month insurance hai. Har order 0.5 hour leta hai aur \$0.50 ka gasoline. Grocers ko cash dete ho. Tumhara outside option: \$6/hour wali dining-hall job.

Q1 — “Explicit costs include...” (Answer: All of these). Explicit cost woh hai jiska cheque tum kaatte ho: web hosting, gasoline, grocers ko paisa — sab cash outflow. Isliye “All of these” sahi hai.

Q2 — “Implicit cost will be...” (Answer: \$6/hour dining-hall wage). \$300 car payment aur \$100 insurance cash mein diye $\Rightarrow$ explicit. Jo \$6/hour tum business chalaane ke liye chhodte ho — woh *implicit* cost hai.

Why examiner asks: explicit vs. implicit ka distinction test karna. **Keywords:** cash outflow, foregone wage, opportunity cost. **Trap:** ek cost ya toh explicit hai *ya* implicit — dono nahi.

PYQ Family — The Fritos Family (Saare 5 terms)

Generic story (parameters P_0, f, S): tumne P_0 mein 5 cases Fritos kharide jo reject ho gaye. Grocer return par har rupaye ke f rupaye refund deta hai (total refund = $f \cdot P_0$). Ek fraternity poore lot ke liye S offer karti hai.

Term	P_0	f	S	Opportunity Cost
2024T1	\$100	0.25	\$55	\$25
2024T2	\$100	0.25	\$55	\$25
2025T1	\$100	0.25	\$55	\$25
2025T2	\$200	0.50	\$125	\$100
Jan 2026	\$200	0.50	\$125	\$100

General solution. P_0 SUNK hai — ignore karo. Frat-sale ki opportunity cost = $f \cdot P_0$; net gain = $S - f \cdot P_0$. Bech do iff $S > f \cdot P_0$.

- 2024T1/T2 & 2025T1: $fP_0 = 0.25 \times 100 = 25 < 55 \Rightarrow$ sell, OC = \$25.
- 2025T2 & Jan 2026: $fP_0 = 0.5 \times 200 = 100 < 125 \Rightarrow$ sell, OC = \$100.
- Sunk cost asked separately = $P_0(1 - f)$; e.g. $100 \times 0.75 = \$75$.

2.3 Fixed Cost, Variable Cost, Average Cost

Decomposition of Total Cost

$$C(Q) = TFC + TVC(Q),$$

yahaan:

- **Fixed cost** TFC — output Q ke saath change nahi hota (factory ka rent, kharidi gayi machinery, manager ki salary, bus ka insurance).
- **Variable cost** $TVC(Q)$ — Q ke saath change hota hai (raw materials, fuel, hourly labour).
- **Quasi-fixed cost** — lagbhag fixed; production start hote hi 0 se ek constant par jump karta hai (e.g. hydro generator ka startup cost). Is course mein hum ise fixed hi maante hain.

2.3.1 Average Costs

$$AC(Q) = \frac{C(Q)}{Q} = \frac{TFC}{Q} + \frac{TVC(Q)}{Q} = AFC(Q) + AVC(Q).$$

Common Mistake — Confusion Alert

- *Fixed total cost* fixed hota hai; par **average fixed cost** Q ke saath *girta* hai, kyunki wahi FC zyada units par spread ho jaata hai. $AFC(Q) = TFC/Q$ ek hyperbola hai.
- *Variable total cost* Q ke saath badhta hai, par **average variable cost** badh, ghat, ya constant reh sakta hai — technology par depend karta hai.

Quick numerical example. $TFC = 500$, $TVC = 5Q$ (constant marginal cost).

$$AFC = \frac{500}{Q}, \quad AVC = \frac{5Q}{Q} = 5, \quad AC = 5 + \frac{500}{Q}.$$

$Q = 10$ par: $AFC = 50$, $AVC = 5$, $AC = 55$. $Q = 100$ par: $AFC = 5$, $AVC = 5$, $AC = 10$.

2.4 Marginal Cost — Ek aur Unit ka Cost

Marginal Cost — Teen Equivalent Definitions

Rudimentary: $MC(Q) = C(Q+1) - C(Q)$.

Discrete: $MC(Q) = \frac{C(Q+h) - C(Q)}{h}$, $h = \pm 1$.

Calculus (most accurate): $MC(Q) = \lim_{h \rightarrow 0} \frac{C(Q+h) - C(Q)}{h} = \frac{dC}{dQ}$.

Marginal Cost = Marginal Variable Cost

Kyunki $C(Q) = TFC + TVC(Q)$ aur TFC constant hai,

$$MC(Q) = \frac{dC}{dQ} = \frac{dTVC}{dQ}.$$

Fixed costs marginal cost ko affect nahi karte.

2.4.1 The MC–AC Relationship (Crucial!)

Numerical intuition. Agar 10 units ka cost Rs 100 hai aur 11th unit ka cost Rs 11 (yaani $MC = 11$), toh naya total 111 hai over 11 units, so $AC = 10.09$. MC (11) purane AC (10) se zyada tha, aur AC badh gaya.

Calculus proof. Maan lo $AC(Q) = C(Q)/Q$. Differentiate:

$$\frac{dAC}{dQ} = \frac{C'(Q) \cdot Q - C(Q)}{Q^2} = \frac{1}{Q} \left(C'(Q) - \frac{C(Q)}{Q} \right) = \frac{1}{Q} (MC(Q) - AC(Q)).$$

The MC–AC Rule

- $MC > AC \Rightarrow AC$ rising.
- $MC < AC \Rightarrow AC$ falling.
- $MC = AC \Rightarrow AC$ apne *minimum* (ya maximum) par.

Geometric fact: MC curve, AC curve ko uske *minimum* par cut karti hai.

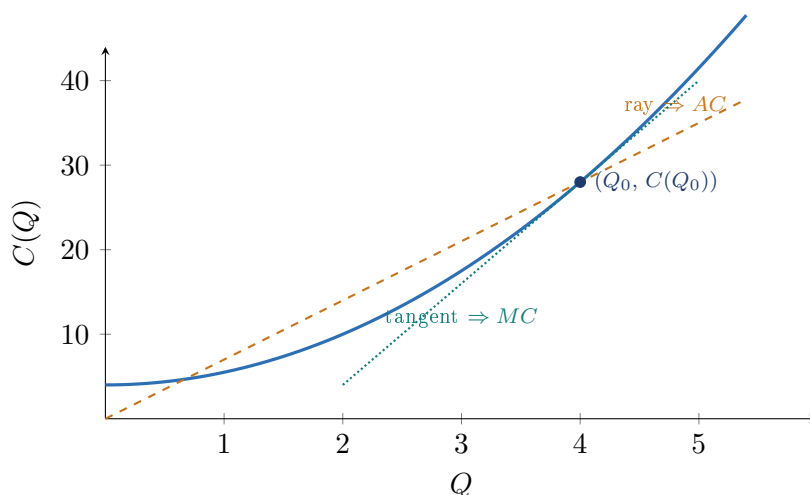
Cricket Analogy

Apne batting average ke baare mein socho. Agar aaj tum apne average se *upar* batting karte ho (tumhara “marginal” score), toh average badh jaata hai. Agar neeche, toh ghat jaata hai. Dono us exact moment par barabar hote hain jab average peak ya trough par hota hai.

2.4.2 Graphical Interpretation of MC and AC

Total cost curve $C(Q)$ par (Q x-axis, C y-axis):

- Kisi point Q_0 par AC = origin se us point tak ki **ray** ka slope.
- Kisi point Q_0 par MC = us point par curve ki **tangent** ka slope.



PYQ Connection — “If $MC = AC$, then AC is...”

Answer: Minimised. Reasoning: $\frac{dAC}{dQ} = \frac{1}{Q}(MC - AC)$. Jab $MC = AC$, derivative = 0 — ek stationary point. Kyunki AC U-shaped hai (pehle girta phir badhta), yeh stationary point ek *minimum* hai. (Appears across 2024–2026 papers.)

2.5 Production Functions — Firm as a Black Box

Definition — Production Function

Ek **production function** inputs ko *maximum feasible output* se map karti hai:

$$Y = f(X_1, X_2, \dots, X_n)$$

yahaan Y output hai (aksar Q likha jaata hai) aur X_i inputs. Hum aam taur par do inputs use karte hain: Labour L aur Capital K , yaani $Y = f(L, K)$.

Capital ke liye letter K historical hai — Marx ki *Das Kapital* (German) se.

Do Concepts of Efficiency

Technical (technological) efficiency: diye gaye input bundle se maximum output produce karna — ek *engineering* concept.

Economic efficiency: target output ko, input prices diye hone par, *minimum cost* par produce karna — ek *managerial* concept.

Production function f already technical efficiency embed kar leti hai. Economic efficiency ke liye humein input prices bhi pata hone chahiye.

2.6 Three Canonical Production Functions

2.6.1 (1) Leontief / Fixed-Proportion

Inputs ko ek fixed ratio mein hi combine karna padta hai (sandwich example: 2 slices bread + 1 spoon butter per sandwich).

$$Y = \min \left\{ \frac{X_1}{a}, \frac{X_2}{b} \right\}.$$

5 bread slices aur 2 spoons butter ke saath ($a = 2, b = 1$): $Y = \min\{5/2, 2/1\} = \min\{2.5, 2\} = 2$ sandwiches. Ek slice bread waste ho jaati hai.

Isoquant. L-shaped (right angles). Saari technical-efficiency corner par hoti hai.

2.6.2 (2) Linear / Perfect Substitutes

Inputs ek fixed exchange ratio mein perfectly substitutable hain:

$$Y = aX_1 + bX_2.$$

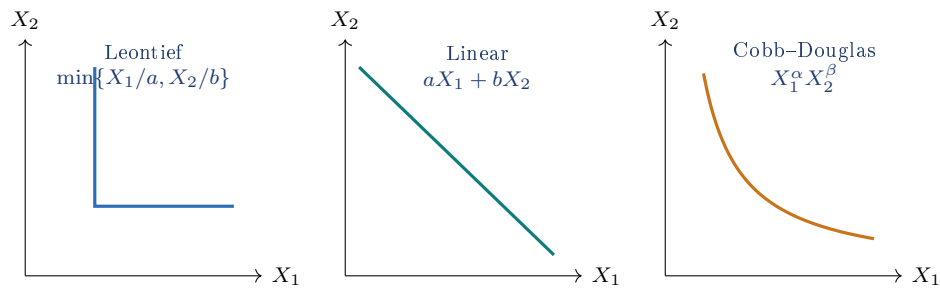
Isoquant. Constant slope $-a/b$ wali ek straight line (X_1 - X_2 space mein).

2.6.3 (3) Cobb–Douglas

The workhorse production function:

$$Y = A X_1^\alpha X_2^\beta.$$

Isoquant. Ek smooth, convex hyperbola-jaisi curve. Jaise $X_1 \rightarrow 0, X_2 \rightarrow \infty$ (aur vice versa) taaki output maintain rahe. Inputs *imperfect* substitutes hain.



Identifying Which Production Function a Question Uses

PYQ language cues:

- “Trade a units of one input for b units of the other, output constant” $\Rightarrow$ **Linear**.
- “Exactly two workers needed per machine” / “fixed proportion” $\Rightarrow$ **Leontief**.
- “ $Y = K^a L^b$ ” ya inputs par koi exponents $\Rightarrow$ **Cobb–Douglas**.

PYQ Family — Production-Function Identification (2024T2 / Jan 2026)

(a) “A firm can trade 3 units of labour for 1 unit of capital keeping output constant.” Constant exchange ratio (MRTS constant) $\Rightarrow$ **Linear** ($Y = aL + bK$). Cobb–Douglas ka MRTS L, K ke saath badalta, Leontief mein kink ke aage trade ho hi nahi sakta.

(b) “Two full-time workers required per piece of machinery.” Fixed proportion $\Rightarrow$ **Leontief**.

(c) “ $q = 4L^{1/2} + 4K$.” $MP_L = \frac{\partial q}{\partial L} = \frac{2}{\sqrt{L}}$ (girta hai, *diminishing*); $MP_K = 4$ (*constant*).

Answer: MP_L decreasing, MP_K constant.

PYQ Family — Cobb–Douglas Returns to Scale (2024T1 / 2025T1)

Q. “The Cobb–Douglas production function $F(K, L) = K^p L^q \dots$ ”

Answer. “Increasing, decreasing, ya constant returns to scale dikhati hai — $p + q$ ki value par depend karta hai.” Reason: $F(\lambda K, \lambda L) = \lambda^{p+q} F(K, L)$, isliye $p + q = 1$ CRS, > 1 IRS,

< 1 DRS. (“Sirf CRS” ya “constant MP ” wale options galat hain.) Jan 2026 mein agar sirf “may exhibit CRS” option ho, toh wahi sahi maano kyunki baaki teen clearly galat hain.

2.7 Isoquants

Definition — Isoquant

Ek **isoquant** input space (e.g. L - K plane) mein woh curve hai jo un saare input bundles ko jodti hai jo *same level of output* dete hain — production function ka ek “level curve”.

Example. $Y = X_1 X_2$.

- $Y = 6$: $X_1 X_2 = 6$ — rectangular hyperbola.
- $Y = 10$: ek aur hyperbola, origin se aur door.

Isoquant map mein aise kayi curves hote hain. Upar wale isoquants higher output ko correspond karte hain.

2.8 Total, Average, and Marginal Product (TP, AP, MP)

Maan lo $Y = f(X)$ output hai jab input X use hota hai (baaki inputs fixed).

Productivity Definitions

$$\begin{aligned}\text{Total product (TP)} &= Y = f(X) \\ \text{Average product (AP)} &= \frac{Y}{X} = \frac{f(X)}{X} \\ \text{Marginal product (MP)} &= \frac{dY}{dX} = f'(X) \quad (\approx f(X+1) - f(X))\end{aligned}$$

2.8.1 Marginal Products with Two Inputs

$$MP_L = \frac{\partial Y}{\partial L}, \quad MP_K = \frac{\partial Y}{\partial K}.$$

Diminishing marginal productivity (assumed): $\frac{\partial MP_L}{\partial L} < 0$, yaani $f_{LL} < 0$.

2.8.2 MP–AP Relationship (Mirrors MC–AC!)

Bilkul wahi calculus jaisa pehle:

$$\frac{dAP}{dX} = \frac{1}{X}(MP - AP).$$

Remember This

- $MP > AP \Rightarrow AP$ rising.
- $MP < AP \Rightarrow AP$ falling.
- $MP = AP \Rightarrow AP$ apne *maximum* par.

Yeh MC–AC rule ka *production-side twin* hai. Dono lagbhag har quiz mein aate hain.

PYQ Family — MP Definitions & AP–MP Relations (2024–2026)

- **Def. of MP_L :** “baaki inputs constant rakhte hue, labour ki ek aur unit lagaane se hone wala extra output.” (Y/L average hota hai, marginal nahi.)
- **Def. of MP_K :** “baaki inputs constant rakhte hue, capital ki ek aur unit se extra output.”
- “ AP decreasing $\Rightarrow MP > AP$ ” — **False** (girte AP par $MP < AP$).
- “ AP increasing $\Rightarrow MP < AP$ ” — **False** (AP rising $\Leftrightarrow MP > AP$).
- “ AP maximum par $\Rightarrow MP = AP$ ” — **True**.

Memory Trick — MP–AP / MC–AC

“The new score pulls the average toward it.” Agar aaj ka latest exam (MP) tumhare current GPA (AP) se upar hai, toh GPA badh jaayega; neeche hai toh gir jaayega; barabar hai toh GPA apne extremum par.

2.9 Forms of Business Organisation (Brief)

Chaar canonical legal forms ka snapshot (PYQ MCQs ke liye relevant):

	Sole Prop.	Partnership	LLP	Corporation
Owners	1 (or family)	2 or more	2 or more	many (shareholders)
Registration	not needed	varies	required	required
Liability	unlimited	unlimited (shared)	limited	limited
Taxation	once (income)	once (income)	once	twice (corp + income)
Life of firm	until owner	until partners	longer	perpetual
Example	local store, taxi	law firm	Sony, Nike	Pepsi, Pfizer, Microsoft, Ford

India mein: add One Person Company (2013), Pvt Ltd, Public Ltd, Section 8 (non-profit), Joint Hindu Family business, aur joint ventures.

2.10 Profit Maximisation — The Marginal Principle

The Marginal Optimality Condition

Q choose karke $\pi(Q) = TR(Q) - TC(Q)$ maximise karne wali firm ko first-order condition milti hai:

$$\frac{d\pi}{dQ} = 0 \iff MR(Q^*) = MC(Q^*)$$

jahaan $MR = dTR/dQ$ aur $MC = dTC/dQ$.

Kyun? Agar $MR > MC$, agli unit cost se zyada revenue laati hai — Q badhao. Agar $MR < MC$, last unit revenue se zyada cost laati hai — Q ghatao. Equality margin par break-even hai.

Discrete Case

Agar Q sirf integer steps mein badal sakta hai, woh Q^* choose karo jahaan

$$MR(Q^*) \geq MC(Q^*) \quad \text{and} \quad MR(Q^* + 1) < MC(Q^* + 1).$$

2.10.1 Detailed Worked Example — Train Operations**Real-World Application**

Tum Indian Railways ke regional manager ho aur cities A aur B ke beech roz kitni trains chalani hain yeh decide karna hai. Daily cost per train = Rs 15 lakh.

Trains n	TC (Rs L)	Total Benefit (Rs L)	Avg. Benefit	Marginal Benefit
0	0	0	—	—
1	15	30	30	30
2	30	48	24	18
3	45	60	20	12
4	60	64	16	4
5	75	66	13.2	2

Marginal Cost = $MC = 15$ lakh (har additional train ka cost 15 L).

Decision rule. Agli train tab chalao jab $MB \geq MC$.

- Train 1: $MB = 30 \geq 15$? Yes. Run.
- Train 2: $MB = 18 \geq 15$? Yes. Run.
- Train 3: $MB = 12 < 15$. Stop.

Optimal number of trains = 2. Saari 6 options par brute-force karne se bhi wahi answer aata hai; marginal approach faster hai aur continuous problems tak generalise hoti hai.

2.10.2 Worked Example — Continuous Profit Maximisation

Ek firm $P = 66$ (price-taker) face karti hai aur total cost $C(Q) = \frac{Q^3}{3} - 10Q^2 + 102Q + (\text{fixed})$. Profit-maximising Q^* nikalo.

Step 1 — MR. $TR = 66Q \Rightarrow MR = 66$.

Step 2 — MC. $MC = \frac{dC}{dQ} = Q^2 - 20Q + 102$.

Step 3 — Set MR = MC.

$$Q^2 - 20Q + 102 = 66 \implies Q^2 - 20Q + 36 = 0.$$

$(Q - 10)^2 = Q^2 - 20Q + 100$, toh $Q^2 - 20Q + 36 = (Q - 10)^2 - 64 = 0$, yaani $(Q - 10)^2 = 64$, i.e. $Q - 10 = \pm 8$. So $Q = 2$ ya $Q = 18$.

Step 4 — Kaunsa root? MC curve U-shape ki hai. $Q = 2$ par MC falling branch par hai (so Q thoda badhaane par MC 66 se neeche aata hai aur profit badhta hai) — yeh profit ka *local minimum* hai. $Q = 18$ par MC rising branch par hai aur MR ko neeche se cross karta hai — yahi profit-maximising quantity hai. $Q^* = 18$.

Second-Order Check

Full second-order condition: $\frac{d^2\pi}{dQ^2} < 0$, yaani $\frac{dMC}{dQ} > 0$ at Q^* . Toh woh root pick karo jahaan **MC rising** hai. Equivalently: jab MR ek horizontal line ho aur MC U-shaped ho, do roots mein se *bada* wala pick karo.

2.11 How to Write This Unit in the Exam**Answer-Writing Strategy**

- **Important keywords:** explicit cost, implicit/opportunity cost, sunk cost, economic profit, TFC , TVC , AFC , AVC , AC , $MC = dTVC/dQ$, isoquant, MP_L , MP_K , returns to scale, $MR = MC$.
- **Diagram suggestions:** (i) total-cost curve par ray (AC) vs. tangent (MC); (ii) teeno isoquants (L-shaped Leontief, straight Linear, convex Cobb–Douglas); (iii) U-shaped MC cutting AC at its minimum. Axes label karna na bhoolo.
- **Numericals mein steps dikhao:** given $\rightarrow$ formula $\rightarrow$ substitution $\rightarrow$ answer. $TC = a + bQ + cQ^2$ se direct read off karo: $TFC = a$, $AVC = b + cQ$, $MC = b + 2cQ$.

Common Mistakes to Avoid

- Sunk cost ko decision mein count karna (Fritos/snack trap) — mat karo.
- Ek cost ko explicit *aur* implicit dono maan lena — woh sirf ek hi hota hai.
- Profit-max mein U-shaped MC ke do roots mein se *falling* branch wala pick karna — hamesha *rising* branch (bada root) lo.
- MC mein fixed cost ghusaa dena — MC par TFC ka koi asar nahi.

2.12 Unit Summary — Quick Revision**Week 2 Quick-Revision Sheet****Costs:**

- Total cost $C(Q) = TFC + TVC(Q)$.
- Explicit (paid out) vs. Implicit (opportunity) vs. Sunk (gone forever).
- Economic Profit = TR – explicit – implicit cost.
- $AFC = TFC/Q$ (girta hai), $AVC = TVC/Q$, $AC = AFC + AVC$.
- $MC = dC/dQ = dTVC/dQ$ (fixed cost vanish!).

Production:

- Teen canonical functions: Leontief (fixed-proportion), Linear (perfect substitutes), Cobb–Douglas.
- Isoquant = inputs ke combinations jo same Y dein.
- AP , MP ka rishta AC , MC ko mirror karta hai.

Optimisation Master Rule: $MR = MC$ at profit-maximising Q^* (with SOC: MC rising

through MR).

Most Important Exam Topics

1. Story mein explicit, implicit, sunk costs identify karna (PYQ favourite!).
2. Ek TC equation se diye gaye Q par AFC , AVC , MC compute karna.
3. Verbal cues se Leontief / Linear / Cobb–Douglas pehchaanna.
4. MC–AC (aur MP–AP) relationship state karna.
5. Profit maximisation: $MR = MC$ with second-order check.

Frequently Asked PYQs

1. Snack-delivery: explicit costs = “All of these”; implicit cost = foregone \$6/hr wage.
2. Fritos: opportunity cost = $f \cdot P_0$ (sunk P_0 ignore); sell iff $S > fP_0$.
3. “Trade 3 labour for 1 capital” $\Rightarrow$ Linear; “2 workers per machine” $\Rightarrow$ Leontief; exponents $\Rightarrow$ Cobb–Douglas.
4. “If $MC = AC$ then AC is minimised.”
5. AP–MP truth/false set (rising $AP \Leftrightarrow MP > AP$).

Expected Long Questions

1. Explicit, implicit aur sunk cost ko define karke ek story (snack/Fritos) ke through illustrate karo; accounting vs. economic profit ka table banao.
2. Teeno canonical production functions, unke isoquants aur MRTS-cues describe karo, diagrams ke saath.
3. Continuous profit maximisation ($MR = MC$) ko ek U-shaped MC example se derive karo, second-order check ke saath ($Q^* = 18$ wala).

Expected Short Questions

1. $MC = dTVC/dQ$ kyun? (fixed cost ka role.)
2. MC–AC rule state karo aur geometric fact batao.
3. $TC = a + bQ + cQ^2$ se TFC , AVC , MC likho.
4. Technical vs. economic efficiency mein ek difference.

One-Night-Before-Exam Revision Notes

- $C = TFC + TVC$; $MC = b + 2cQ$, $AVC = b + cQ$, $AFC = a/Q$ (for $TC = a + bQ + cQ^2$).
- $MC \gtrless AC \Rightarrow AC$ rising/falling; equal $\Rightarrow AC$ minimum.
- Leontief $\min\{\cdot\}$ L-shape; Linear straight; Cobb–Douglas convex.
- $MR = MC$, rising branch; bada root pick karo.
- Sunk cost ignore; implicit cost hamesha count.

Quick Formula Sheet (Week 2)

$$TC = TFC + TVC, \quad AC = \frac{TC}{Q}, \quad AFC = \frac{TFC}{Q}, \quad AVC = \frac{TVC}{Q}, \quad MC = \frac{dTC}{dQ} = \frac{dTVC}{dQ}.$$

$$\text{For } TC = a + bq + cq^2 : \quad MC = b + 2cq, \quad AVC = b + cq, \quad AC = \frac{a}{q} + b + cq.$$

$$MP_L = \frac{\partial Q}{\partial L}, \quad MP_K = \frac{\partial Q}{\partial K}, \quad AP_L = \frac{Q}{L}, \quad \frac{dAC}{dQ} = \frac{1}{Q}(MC - AC), \quad \frac{dAP}{dX} = \frac{1}{X}(MP - AP).$$

$$\text{Profit max: } MR = MC \text{ (rising branch),} \quad \text{Cobb-Douglas RTS: } \alpha + \beta \gtrless 1.$$

Practice Questions — Week 2

- $TC(Q) = 200 + 50Q + Q^2$. TFC , AVC at $Q = 10$, AC at $Q = 10$, aur MC at $Q = 10$ nikalo.
- Ek bakery owner Rs 50,000/month ki salary chhod kar bakery chalata hai. Monthly explicit costs Rs 1,20,000 hain. Revenue Rs 1,60,000.
 - Accounting profit?
 - Implicit cost?
 - Economic profit?
- Match karo production function se: “perfect substitutes”, “fixed ratio”, “ $Y = K^{0.5}L^{0.5}$ ”.
- $Y = 4K^{1/2} + 4L$ ke liye MP_K aur MP_L nikalo. Batao har ek constant hai ya declining.
- Agar MC barabar AC ho, toh us Q par AC ke baare mein kya conclude hota hai?

Unit 3

MRTS, Cost Minimisation, Returns to Scale, and Supply

3.1 Recap and Roadmap

Unit 2 mein humne production aur cost ka base banaya. Ab Unit 3 us base se *supply curve* tak ka bridge banata hai. Naye key ideas:

- **Marginal Rate of Technical Substitution (MRTS)** — isoquant ka slope.
- **Cost minimisation** — target output ko sabse sasta produce karne wale inputs choose karna.
- Leontief, Linear, aur Cobb–Douglas technologies ke liye cost function $C(Q)$.
- **Returns to scale** aur short run vs. long run.
- Marginal cost curve se **supply function** derive karna.

3.2 Marginal Rate of Technical Substitution (MRTS)

MRTS — Definition

Labour-for-capital ka **MRTS** woh capital ki maatra hai jo firm har additional unit labour ke badle chhod sakti hai, output ko constant rakhte hue:

$$MRTS_{L,K} = - \frac{dK}{dL} \Big|_{Y=\text{const}}.$$

Yeh kisi point par isoquant ke slope ka (absolute value) hai.

3.2.1 MRTS Formula ko Derive Karna

$Y = f(L, K)$ ka total differential lo, Y ko fixed rakhte hue:

$$dY = f_L dL + f_K dK = 0 \implies \frac{dK}{dL} \Big|_{Y=\text{const}} = - \frac{f_L}{f_K} = - \frac{MP_L}{MP_K}.$$

Isliye

$$MRTS_{L,K} = \frac{MP_L}{MP_K} \quad (\text{magnitudes; geometric slope negative}).$$

Intuition

Agar tum capital ki ek unit chhodte ho, toh MP_K units output kho dete ho. Use recover karne ke liye tumhe $\frac{MP_K}{MP_L}$ units labour chahiye. Isliye capital ki ek unit MP_K/MP_L units labour se “exchange” hoti hai — aur labour ki ek unit MP_L/MP_K units capital se. Direction ka dhyaan rakho!

Worked numerical. $Y = LK$ (Cobb–Douglas, exponents 1,1). $MP_L = K$, $MP_K = L$. So $MRTS_{L,K} = K/L$.

- $(L, K) = (3, 4)$ par: $MRTS = 4/3$. 1 unit K chhodne ke liye $3/4$ unit L chahiye.
- $(L, K) = (2, 6)$ par: $MRTS = 6/2 = 3$. Zyada capital intensity $\Rightarrow$ labour margin par “more powerful”.

Convexity. Cobb–Douglas isoquants convex hote hain (origin ki taraf jhuke). Isoquant par right ki taraf jaate hue L badhta hai aur K ghatta hai, toh $MRTS$ *ghatta* hai — yeh **diminishing MRTS** hai.

3.2.2 MRTS for the Three Canonical Technologies

Technology	Form	MRTS
Linear (perfect substitutes)	$Y = aL + bK$	a/b (constant)
Leontief (fixed proportion)	$Y = \min\{L/a, K/b\}$	0 or ∞ (kinked)
Cobb–Douglas	$Y = AL^\alpha K^\beta$	$\frac{\alpha K}{\beta L}$ (diminishing)

Exam Tip

PYQ: “A firm can always trade 3 units of labour for 1 unit of capital and keep output constant.”
Constant exchange ratio $\Rightarrow$ **Linear** production function.

3.3 Cost Minimisation — The Manager’s Problem

Definition — Cost Minimisation

Input prices w (labour wage) aur r (capital rental) ke saath cost-minimisation problem:

$$\min_{L, K} wL + rK \quad \text{subject to} \quad f(L, K) \geq Y_0.$$

Isocost line: $wL + rK = C$ — woh sab (L, K) bundles jinka same expenditure hai. Slope $= -w/r$. Jaise C badhta hai, isocost line bahar ki taraf parallel shift karti hai.

3.3.1 The Tangency Condition (Interior Solution)

Cost-minimising interior point (L^*, K^*) par, isoquant isocost line ko tangent hoti hai:

$$MRTS_{L,K} = \frac{MP_L}{MP_K} = \frac{w}{r}.$$

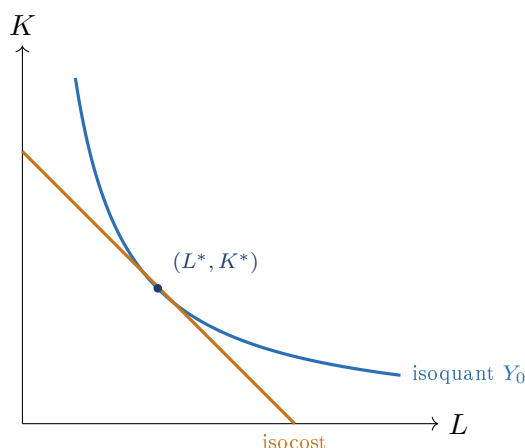
Equimarginal Principle

Tangency condition ko rearrange karo:

$$\frac{MP_L}{w} = \frac{MP_K}{r}.$$

Har input par kharch kiye gaye *rupaye per marginal output* barabar hone chahiye. Agar $MP_L/w > MP_K/r$, toh labour se “zyada bang per buck” milta hai — zyada labour, kam capital use karo, jab tak ratios barabar na ho jaayein.

3.3.2 Visualising Cost Minimisation



3.3.3 Cost Minimisation for Leontief (Corner)

$Y = \min\{L/a, K/b\}$ ke L-shaped isoquants hote hain. Technical efficiency par $L/a = K/b$, yaani $L = aY_0$ aur $K = bY_0$. Cost function:

$$C(Y_0) = w \cdot aY_0 + r \cdot bY_0 = (wa + rb) Y_0.$$

3.3.4 Cost Minimisation for Linear (Often Corner)

$Y = aL + bK$. $MRTS = a/b$ constant; isoquants straight lines.

- $w/r < a/b$ (labour apni productivity ratio ke relative “sasta”): sirf labour use karo, $L = Y_0/a$, $K = 0$.
- $w/r > a/b$: sirf capital use karo, $K = Y_0/b$, $L = 0$.
- $w/r = a/b$: koi bhi combination chalega.

PYQ Recipe for Linear Cost-Minimisation

$Y = aL + bK$ ke saath, prices w, r :

1. $MP_L/w = a/w$ aur $MP_K/r = b/r$ compute karo.
2. Jiska ratio bada ho, sirf usi input ko use karo — woh per rupaya sabse zyada output deta hai.
3. Constraint $aL^* = Y_0$ (ya $bK^* = Y_0$) se L^* ya K^* nikalo.

Example. $Y = 10L + 2K$, $w = 5$, $r = 2$, $Y_0 = 200$. $MP_L/w = 2$, $MP_K/r = 1$. Labour wins. $L^* = 20$, $K^* = 0$. Cost = \$100.

3.3.5 Cost Minimisation for Cobb–Douglas (Interior)

$Y = AL^\alpha K^\beta$. $MRTS = \frac{\alpha K}{\beta L}$. Set $MRTS = w/r$:

$$\frac{\alpha K}{\beta L} = \frac{w}{r} \implies K = \frac{\beta w}{\alpha r} L.$$

Ise constraint $Y_0 = AL^\alpha K^\beta$ mein substitute karke L^* nikalo, phir K^* .

Concrete example. $Q = 50\sqrt{LK}$, $w = 5$, $r = 20$, $Q_0 = 1000$. Yahaan $\alpha = \beta = 1/2$. Tangency: $\frac{K}{L} = \frac{w}{r} = \frac{5}{20} = \frac{1}{4}$, so $K = L/4$. Constraint mein daalo:

$$1000 = 50\sqrt{L \cdot L/4} = 50 \cdot \frac{L}{2} = 25L \implies L^* = 40, K^* = 10.$$

$$C^* = 5 \cdot 40 + 20 \cdot 10 = \$400.$$

PYQ Family — Cobb–Douglas Cost Minimisation (2024–2026)

General recipe: (1) tangency $\frac{MP_L}{MP_K} = \frac{w}{r}$; (2) K ko L ke terms mein solve; (3) constraint mein substitute karke L^*, K^* ; (4) $C^* = wL^* + rK^*$.

Variant	L^*	K^*	C^*
$q = 10L^{1/2}K^{1/2}$, $w = 20, r = 80, Q_0 = 100$	20	5	\$800
$Q = 50\sqrt{LK}$, $w = 5, r = 20, Q_0 = 1000$ (2025T1)	40	10	\$400
$Q = 50\sqrt{LK}$, $w = 20, r = 5, Q_0 = 1000$ (2025T2)	10	40	\$400
$q = 100KL$, $r = 1200, w = 300, q_0 = 10000$ (2024T1)	20	5	\$12000

Quiz Trap (2025T1 vs 2025T2): prices swap hone se (L^*, K^*) mirror ho jaate hain — (40, 10) vs. (10, 40). *Sasta input zyada intensively use hota hai.* For $q = 100KL$: $K/L = w/r = 300/1200 = 0.25$ (often a sub-question).

PYQ Family — Chair Manufacturer (Perfect Substitute / Linear)

Story. Ek chair 4 hours labour *ya* machinery se banti hai (perfect substitutes, 1 : 1, $MRTS = 1$). Sasta input dominate karega.

- **2024T1** ($w = \$30$ labour, $r = \$15$ machinery): machinery per dollar double chair deti hai $\Rightarrow$ use only machinery: $L^* = 0$, $K^* = 4$ (per chair).
- **2024T2 & Jan 2026** ($w = \$15$, $r = \$30$): labour sasta $\Rightarrow L^* = 4$, $K^* = 0$ (per chair).

Universal rule: $L^* > 0, K^* = 0 \iff \frac{MP_L}{w} > \frac{MP_K}{r} \iff \frac{a}{w} > \frac{b}{r}$; warna corner flip; equality par koi bhi mix.

PYQ Family — Linear Production & Optimal Inputs

(2025T1) $Q = 10L + 2K$, $w = 5$, $r = 2$, $Q_0 = 200$. $MP_L/w = 2 > MP_K/r = 1 \Rightarrow$ only labour: $L^* = 20$, $K^* = 0$, $TC = \$100$.

(2025T2) $Q = 5L + 2K$, $w = 5$, $r = 2$, $Q_0 = 200$. $MP_L/w = 1 = MP_K/r = 1$ — **tie!** Koi bhi (L, K) jo $5L + 2K = 200$ satisfy kare, cost minimise karta hai. Answer key $K^* = 100, L^* = 0$ deta hai, par $TC = \$200$ *uniquely* fixed hai ((40, 0), (20, 50) sab same cost).

3.4 The Total Cost Function

Cost-minimisation problem ek by-product ke roop mein long-run total cost function $C(Q)$ deta hai: har output level Q ko produce karne ka minimum total expenditure.

- **Leontief:** $C(Q) = (wa + rb)Q$ — Q mein linear.
- **Linear (corner):** $C(Q) = \min\{w/a, r/b\} \cdot Q$ — linear.
- **Cobb–Douglas** $Y = L^{1/2}K^{1/2}$, $w = 5, r = 20$: $C(Q) = 2\sqrt{wr} \cdot Q = 20Q$ ($2\sqrt{5 \cdot 20} = 20$). Bhi linear!

In teeno constant-returns cases mein $C(Q)$ linear hai aur average cost constant.

3.5 Returns to Scale

Returns to Scale

Production function f aur scale factor $\lambda > 1$ ke liye:

- **Constant returns (CRS):** $f(\lambda L, \lambda K) = \lambda f(L, K)$.
- **Increasing returns (IRS):** $f(\lambda L, \lambda K) > \lambda f(L, K)$.
- **Decreasing returns (DRS):** $f(\lambda L, \lambda K) < \lambda f(L, K)$.

3.5.1 Returns to Scale for Cobb–Douglas

$Y = AL^\alpha K^\beta$. Toh $f(\lambda L, \lambda K) = A(\lambda L)^\alpha (\lambda K)^\beta = \lambda^{\alpha+\beta} AL^\alpha K^\beta = \lambda^{\alpha+\beta} Y$.

Cobb–Douglas Returns

$\alpha + \beta = 1$: constant returns to scale.
 $\alpha + \beta > 1$: increasing returns.
 $\alpha + \beta < 1$: decreasing returns.

Common Mistake

Returns-to-scale tab measure hota hai jab *saare* inputs proportionally scale up hon. Yeh ek single input ke *marginal product* se alag hai (jo baaki fixed rakhta hai). Ek Cobb–Douglas function CRS ($\alpha + \beta = 1$) ho sakta hai aur phir bhi diminishing MP_L ($\alpha < 1$) dikha sakta hai.

3.5.2 Linear and Leontief Returns

Linear ($Y = aL + bK$) aur Leontief ($Y = \min\{L/a, K/b\}$) dono direct substitution se **CRS** dikhate hain.

3.5.3 Economies and Diseconomies of Scale (Cost Side)

- **Economies of scale** — Q badhne par LRAC girta hai. Aam taur par IRS se juda.
- **Diseconomies of scale** — LRAC badhta hai. Aam taur par DRS se juda.
- **Constant** — LRAC constant; CRS.

3.6 Short Run vs. Long Run

Definition

Long run (LR) — itna lamba period ki firm saare inputs vary kar sake. **Short run (SR)** — woh period jismein kam-se-kam ek input (aam taur par capital) fixed ho. SR aur LR calendar concepts nahi hain; industry par depend karte hain. “Do mahine” ek roadside vendor ke liye LR ho sakta hai aur aircraft manufacturer ke liye SR.

3.6.1 Why $SRTC \geq LRTC$

Long run mein firm L aur K dono optimise karti hai. Short run mein K ek $\bar{K}$ par atki hai, sirf L adjust ho sakta hai. Isliye

$$SRTC(Q | \bar{K}) \geq LRTC(Q),$$

equality sirf us Q par jahaan $\bar{K}$ long-run optimal capital level hai.

3.6.2 Short-Run Marginal Cost in Terms of MPL

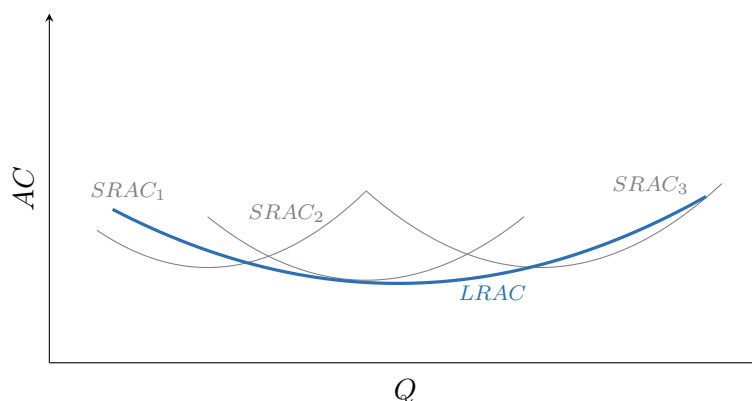
Short run mein sirf labour vary hota hai. Ek aur unit output ke liye firm ko $1/MP_L$ extra labour chahiye, har ek w ka. Isliye

$$MC_{SR} = \frac{w}{MP_L}.$$

Diminishing $MP_L \Rightarrow$ rising MC_{SR} .

3.6.3 Envelope Property

LRAC curve saare possible SRAC curves ka **lower envelope** hai (har $\bar{K}$ ke liye ek). Har Q par LRAC us SRAC ko touch karti hai jiska $\bar{K}$ us Q ke liye long-run optimum hai.



3.7 From Cost Curves to Supply Function

Definition

Supply function $Q^S(P)$ batata hai ki firm har market price P par kitni quantity supply karne ko taiyaar hai. Hum ise cost minimisation + profit maximisation se derive karte hain.

3.7.1 Step-by-Step Derivation (Price-Taker)

Ek price-taker firm P given face karti hai. Profit maximisation: $P = MC(Q)$.

(i) **Sirf upward-sloping part of MC use karo.** Agar MC U-shaped hai, toh $P = MC(Q)$ ke do solutions hote hain. Downward-sloping solution profit ka local minimum hai; sirf upward-sloping piece maximum deta hai.

(ii) **AVC ke upar raho — shutdown rule.** Agar $P < AVC(Q^*)$, firm apna variable cost bhi cover nahi kar paati. Better hai shut down kar do ($Q = 0$).

Supply Function (Short-Run)

Ek price-taking firm ka (short-run) supply curve:

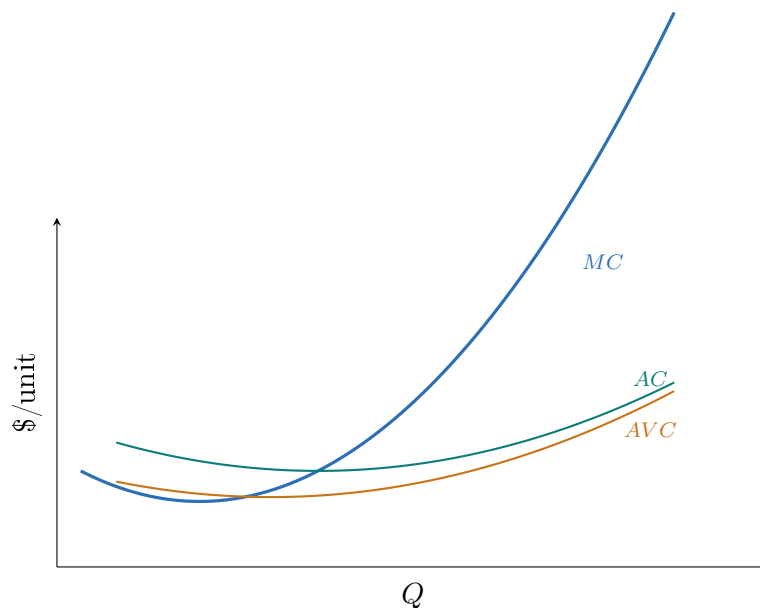
$$Q^S(P) = \begin{cases} \text{solution to } P = MC(Q) \text{ on rising branch,} & P \geq \min AVC, \\ 0, & P < \min AVC. \end{cases}$$

Supply curve, MC curve ka woh part hai jo AVC curve ke *upar* hai.

3.7.2 Why Fixed Cost is Irrelevant to the Shutdown Decision

Short run mein fixed cost *sunk* hai; produce karo ya na karo, dena hi hai. Isliye operating decision revenue ko *variable cost* se compare karta hai, fixed cost se nahi.

3.7.3 Illustrative Diagram — MC, AVC, AC



3.8 Detailed PYQ-Style Walkthrough — $TC = 200 + 50Q + Q^2$

Given. $TC(Q) = 200 + 50Q + Q^2$. Identify:

$$TFC = 200, \quad TVC = 50Q + Q^2, \quad AVC = 50 + Q, \quad AFC = \frac{200}{Q},$$

$$AC = 50 + Q + \frac{200}{Q}, \quad MC = 50 + 2Q.$$

Common PYQ sub-questions:

- AFC at $Q = 100$: $200/100 = 2$.
- AVC at $Q = 25$: $50 + 25 = 75$.
- MC at $Q = 25$: $50 + 2 \cdot 25 = 100$.
- AC ka minimum: $AC'(Q) = 0$ ya $MC = AC$ se solve karo.

PYQ Family — Quadratic & Linear Short-Run Cost

Master read-off from $TC = a + bq + cq^2$: $TFC = a$, $TVC = bq + cq^2$, $AFC = a/q$, $AVC = b + cq$, $AC = a/q + b + cq$, $MC = b + 2cq$.

Variant	AFC	AVC	MC
$TC = 100 + 25q + q^2$ (2024T1)	1 at $q=100$	50 at $q=25$	75 at $q=25$
$TC = 200 + 50q + q^2$ (2025T1)	2 at $q=100$	75 at $q=25$	100 at $q=25$
$TC = 300 + 50q + q^2$ (2025T2)	6 at $q=50$	100 at $q=50$	150 at $q=50$

Linear TC (Jan 2026, 2024T2): $TC = 200 + 55q \Rightarrow TVC = 55q$, $AVC = 55$ (constant), fixed cost = 200.

PYQ Family — MRTS / MP Numeric (Jan 2026 Q19)

Q. Chip production mein $MP_L = 50$ kg/hour; $MRTS_{L,K} = 1/4$. MP_K nikalo.

Concept. $MRTS_{L,K} = \frac{MP_L}{MP_K} \Rightarrow MP_K = \frac{MP_L}{MRTS} = \frac{50}{1/4} = 200$ kg/hour.

Intuition. 1 hour capital chhodne par 4 hours labour free hote hain (output constant), toh capital margin par 4 guna productive: $MP_K = 4 \cdot MP_L = 200$.

3.9 How to Write This Unit in the Exam

Answer-Writing Strategy

- **Keywords:** $MRTS = MP_L/MP_K$, isocost, tangency, equimarginal principle ($MP_L/w = MP_K/r$), corner vs. interior solution, returns to scale, envelope, shutdown rule ($P < \min AVC$), supply = MC above AVC.
- **Diagram suggestions:** (i) isoquant–isocost tangency; (ii) SRAC family with LRAC lower envelope; (iii) U-shaped MC with AVC & AC (supply = MC above min AVC).
- **Cost-min numericals:** hamesha (1) tangency likho, (2) input ratio solve, (3) constraint substitute, (4) C^* compute. Linear/Leontief mein corner check karo.

Common Mistakes to Avoid

- Returns-to-scale ko marginal product se mila dena — alag concepts.
- Supply mein MC ki *falling* branch use kar lena — sirf rising branch.
- Shutdown ke liye $P < \min AC$ likh dena — sahi rule $P < \min AVC$ hai.
- Linear cost-min mein interior solution dhoondhna — aksar *corner* hota hai.

3.10 Unit Summary — Quick Revision

Week 3 Quick-Revision Sheet

MRTS: $MRTS_{L,K} = MP_L/MP_K$ (isoquant slope ka magnitude).

Cost Min: tangency $\Rightarrow MP_L/w = MP_K/r$ (equimarginal principle).

Recipes:

- Leontief: $L^* = aY_0$, $K^* = bY_0$, $C(Y_0) = (wa + rb)Y_0$.
- Linear: corner — cheaper-per-output input use karo.
- Cobb–Douglas $Y = L^\alpha K^\beta$: tangency $\Rightarrow K/L = (\beta w)/(\alpha r)$.

Returns to Scale (CD): $\alpha + \beta = 1$ CRS, > 1 IRS, < 1 DRS.

SR vs LR: $SRTC \geq LRTC$; LRAC = lower envelope of SRACs; $MC_{SR} = w/MP_L$.

Supply: MC ka upward-sloping part, above min AVC. Neeche — shut down.

Most Important Exam Topics

1. Diye gaye $Y(L, K)$ se MRTS compute karna.
2. w, r, Y_0 diye hone par linear / Leontief / Cobb–Douglas ke liye L^*, K^* .
3. Returns to scale identify karna.
4. Shutdown rule pehchaanna.
5. Short run mein $MC = w/MP_L$ compute karna.

Frequently Asked PYQs

1. Cobb–Douglas cost-min variants (prices swap $\Rightarrow$ mirror inputs).
2. Chair manufacturer perfect-substitute corner solution.
3. Linear production tie case ($Q = 5L + 2K$, $TC = \$200$ unique).
4. MP_K from MRTS aur $MP_L (= 200)$.
5. Quadratic/linear TC se AFC, AVC, MC read-off.

Expected Long Questions

1. Equimarginal principle ke saath cost-minimisation derive karo; isoquant–isocost tangency diagram banao; Cobb–Douglas numerical solve karo.
2. MC curve se short-run supply function derive karo; shutdown rule aur “fixed cost irrelevant” explain karo (MC/AVC/AC diagram ke saath).
3. Returns to scale (CRS/IRS/DRS) define karo, Cobb–Douglas par $\alpha + \beta$ test derive karo, aur economies/diseconomies of scale se link karo.

Expected Short Questions

1. MRTS define karo aur teeno technologies ke liye value likho.
2. Equimarginal principle state karo.
3. Shutdown price kya hai aur kyun?
4. $MC_{SR} = w/MP_L$ derive karo.

One-Night-Before-Exam Revision Notes

- Tangency: $MP_L/w = MP_K/r$; CD: $K/L = (\beta w)/(\alpha r)$.
- Linear/Leontief cost-min $\Rightarrow$ corner; CD $\Rightarrow$ interior.
- RTS (CD): $\alpha + \beta \gtrless 1$.
- Supply = rising MC above min AVC ; shutdown at $P < \min AVC$.
- $MC_{SR} = w/MP_L$; diminishing $MP_L \Rightarrow$ rising MC.

Quick Formula Sheet (Week 3)

$$MRTS_{L,K} = \frac{MP_L}{MP_K}, \quad \text{Cost-min: } \frac{MP_L}{w} = \frac{MP_K}{r}, \quad \text{Isocost slope} = -\frac{w}{r}.$$

$$\text{Leontief } C = (wa + rb)Y_0, \quad \text{CD } K = \frac{\beta w}{\alpha r}L, \quad MC_{SR} = \frac{w}{MP_L}, \quad \text{Shutdown: } P < \min AVC.$$

Practice Questions — Week 3

1. $Y = 4L^{1/2}K^{1/2}$, $w = 4$, $r = 16$, $Y_0 = 80$. L^* , K^* , C^* nikalo.
2. $Y = 3L + 4K$, $w = 2$, $r = 5$. $Y_0 = 60$ ke liye cost-minimising combination batao.
3. $Y = 10L^{0.4}K^{0.3}$ ke returns to scale state karo.
4. Ek firm ke SR mein $w = 10$ aur $MP_L = 4$. MC compute karo.
5. $TC = 100 + 10Q$ diya hai — firm ka supply curve kya hai, aur kis P par shut down karna chahiye?

Unit 4

Short-/Long-Run Costs and Consumer Theory

4.1 Recap and Roadmap

Unit 4 do kaam karta hai: pehle cost-curve ki kahani close karta hai (short-run vs. long-run AC & MC), aur phir ek bilkul naya chapter kholta hai — **consumer behaviour**. Hum first principles se banaayenge:

- Consumption sets, budget sets, budget lines.
- Preferences (rationality, continuity, monotonicity, convexity).
- Utility functions — preferences ka numerical representation.
- Indifference curves aur Marginal Rate of Substitution (MRS).

4.2 Short-Run vs. Long-Run Costs — Final Pieces

Long run mein (constant-returns technology), long-run total cost (LRTC) linear hota hai, LRAC constant, aur LRMC = LRAC har jagah. Short run mein capital ek $\bar{K}$ par atki hai.

Remember This

- SRTC ka cost-axis par positive intercept $r\bar{K}$ hota hai = short-run fixed cost.
- $SRTC \geq LRTC$, equality sirf us Q par jahaan $\bar{K}$ long-run optimal hai.
- Bada $\bar{K} \Rightarrow$ zyada fixed cost aur (aam taur par) zyada MP_L , toh $MC_{SR} = w/MP_L$ kam hota hai.
- LRAC saare SRAC curves ka *lower envelope* hai.

4.2.1 Numerical Slope Insight

Yaad karo $MC_{SR} = w/MP_L$. Agar $\bar{K}$ bada hai, capital labour ko complement karta hai (MP_L zyada), toh SRTC curve *flatter* hoti hai. Agar $\bar{K}$ chhota hai, MP_L kam hai aur SRTC zyada steeply climb karti hai.

4.3 Building Blocks of Consumer Theory

Poori picture, order mein:

1. Consumption Possibility Set (sab kuch imaginable).
2. Consumption Feasibility Set (jo actually attainable hai).
3. Preferences over bundles.
4. Behavioural assumptions.
5. Utility function as numerical representation.

4.3.1 Consumption Bundles and Sets

Ek **bundle** ek n -tuple $x = (x_1, x_2, \dots, x_n)$ hai jahaan x_i good i ki quantity hai. Graphs feasible banane ke liye hum $n = 2$ tak restrict karte hain.

Consumption Possibility Set X . Saare conceivable bundles. Non-negative orthant tak restricted: $X = \mathbb{R}_+^n = \{x : x_i \geq 0\}$. Divisibility allowed: aadha litre doodh valid hai.

Consumption Feasibility Set Y (Budget Set). Woh bundles jo consumer prices aur income diye hone par actually afford kar sakta hai.

4.3.2 Budget Constraint — The Workhorse

Prices P_1, P_2 aur income I ke saath, consumer bundle (x_1, x_2) afford kar sakta hai agar

$$P_1x_1 + P_2x_2 \leq I.$$

Budget line equality hai $P_1x_1 + P_2x_2 = I$:

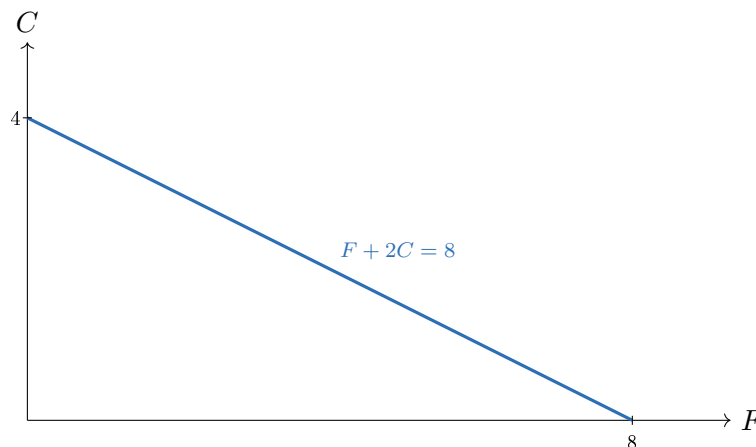
$$x_2 = \frac{I}{P_2} - \frac{P_1}{P_2}x_1.$$

Budget Line — Key Features

- Vertical intercept ($x_1 = 0$): $x_2 = I/P_2$ (saara income good 2 par).
- Horizontal intercept ($x_2 = 0$): $x_1 = I/P_1$ (saara good 1 par).
- Slope: $-P_1/P_2$ (prices ka negative ratio). Magnitude = good 1 ka good 2 ke liye exchange rate: 1 unit good 1 chhodo, P_1 rupaye bacho, jo P_1/P_2 units good 2 kharidte hain.

4.3.3 Robinson Crusoe — Time as Income

Crusoe ke paas roz 8 ghante hain. 1 ghanta = 1 unit fish; 2 ghante = 1 unit coconut. Toh constraint $1 \cdot F + 2 \cdot C \leq 8$ time-budget form mein budget line hai. Vertical intercept (sirf coconut): $C = 4$. Horizontal (sirf fish): $F = 8$. Slope: $-1/2$.



4.4 Shifts and Rotations of the Budget Line

Effects of Parameter Changes

- **Income I changes:** dono intercepts proportionally change; slope same. *Parallel shift.*
- **Price P_1 changes:** horizontal intercept I/P_1 change; vertical same; slope change (vertical intercept ke around rotate).
- **Price P_2 changes:** vertical intercept I/P_2 change; horizontal same; slope change (horizontal intercept ke around rotate).

Worked example (Jan 2026 PYQ). Prices $P_C = 20$, $P_F = 10$, income $I = 1000$, fish y-axis par (so good 1 = coconut, good 2 = fish).

- Intercepts: coconut-only = $1000/20 = 50$; fish-only = $1000/10 = 100$.
- Budget line slope (fish on y): $-P_C/P_F = -20/10 = -2$.
- Agar P_C aadha hokar 10: naya slope = $-10/10 = -1$; naye intercepts 100, 100.
- Agar income aadha (prices same): slope same (-2), dono intercepts halve (25, 50).

PYQ Family — Budget-Line Comprehension (Robin / Coconut-Fish)

$$P_C \cdot C + P_F \cdot F = I; \quad F = \frac{I}{P_F} - \frac{P_C}{P_F} C.$$

- Slope = $-P_C/P_F = -20/10 = -2$. • X-int (coconut) = $I/P_C = 50$; Y-int (fish) = $I/P_F = 100$.
- P_C halved to 10: slope = -1 ; X-int = 100, Y-int unchanged = 100.
- Income halved, prices same: slope same; both intercepts halved (25, 50). **Mnemonic:** income $\uparrow$ = parallel shift out; $P_x \uparrow$ = x-intercept andar, y-pivot; $P_y \uparrow$ = y-intercept andar, x-pivot.

4.5 Preferences — Rationality and Continuity

Hum $\succeq$ symbol use karte hain “at least as good as” ke liye. $x \succeq y$ padho “bundle x weakly preferred to y .”

Rationality Axioms

- **Completeness:** kisi bhi x, y ke liye, ya $x \succeq y$, ya $y \succeq x$, ya dono ($x \sim y$, indifferent).
- **Transitivity:** agar $x \succeq y$ aur $y \succeq z$, toh $x \succeq z$.
- **Reflexivity:** $x \succeq x$.

Ek preference relation **rational** hai agar yeh teeno satisfy kare.

Rationality kyun assume karein? Iske bina hum bundles ko consistently rank nahi kar sakte, aur choice ki koi meaningful theory possible nahi.

4.5.1 Continuity

Continuity

Agar $x_n \rightarrow x$ aur $y_n \rightarrow y$ aise sequences hain jinmein har n ke liye $x_n \succeq y_n$, toh $x \succeq y$. Loosely: preference mein koi achaanak jump nahi; nearby bundles ki utility nearby hoti hai.

4.5.2 The Existence of Utility Functions

Debreu's Theorem (Informally)

Agar $\succeq$ consumption set par rational aur continuous hai, toh ek continuous utility function $u : X \rightarrow \mathbb{R}$ exist karta hai jaisa ki

$$x \succeq y \iff u(x) \geq u(y).$$

Utility is ordinal. Sirf order matter karta hai — absolute number nahi. Koi bhi strictly increasing transformation $g(u(x))$ (e.g. $u + 100$, $3u$, u^3 , $\ln u$) bilkul same preferences represent karta hai. Hum “Bundle A ki happiness Bundle B ki double hai” nahi keh sakte — ordinal utility mein iska koi matlab nahi.

4.6 Behavioural Assumptions on Preferences

4.6.1 (Local) Non-Satiation

Local Non-Satiation

Kisi bhi bundle x aur kisi bhi $\varepsilon > 0$ ke liye, ek bundle y exist karta hai jismein $\|y - x\| < \varepsilon$ aur $y \succ x$. Plain: kisi bhi bundle ke kisi bhi tiny neighbourhood mein ek strictly preferred bundle hota hai. Koi bundle “bliss point” nahi.

4.6.2 Monotonicity (“More is Better”)

Monotonicity

Agar $y \geq x$ componentwise aur $y \neq x$, toh $y \succ x$.

Relationship: Monotonicity $\Rightarrow$ Local non-satiation (bas “more” direction mein thoda move karo). Ulta sach nahi: “less is better” (ajeeb par coherent) local non-satiation satisfy kar sakta hai bina monotonicity ke.

4.6.3 Consequences for Indifference Curves

Remember This

Monotonicity ke under, indifference curves zaroori hain:

1. **Thin** (regions nahi). Do bundles same utility nahi de sakte agar ek mein har good zyada ho.
2. **Downward sloping**. Upward segment ka matlab dono goods zyada wale bundle ki same utility — contradiction.
3. **Non-intersecting**. Crossing transitivity ke saath contradiction deta hai.

4.7 Convexity of Preferences

Convexity

Weak convexity: agar $x \succeq y$, toh $\lambda \in [0, 1]$ ke liye $\lambda x + (1 - \lambda)y \succeq y$.

Strict convexity: wahi, par $\lambda \in (0, 1)$ ke liye strict $\succ$.

Economic interpretation. Balanced bundles extreme bundles se preferred hote hain. Agar tumhe apple aur banana dono pasand hain, toh dono ka mix sirf ek wale basket se kam-se-kam utna hi accha hai.

4.7.1 Convexity and the Shape of Indifference Curves

- **Convex preferences** $\Rightarrow$ ICs origin ki taraf jhuke (upper-contour set convex).
- **Linear preferences** (perfect substitutes) $\Rightarrow$ straight ICs (weakly convex, strictly nahi).
- **Concave preferences** $\Rightarrow$ “bowed out” ICs: extreme bundles mixes se preferred (unusual: e.g. “curd aur chocolate, par saath mein nahi”).

4.8 The Marginal Rate of Substitution (MRS)

MRS — Definition

Good 1 ka good 2 ke liye **MRS** woh good 2 ki maatra hai jo consumer ko good 1 ki ek unit ki kami compensate karne ke liye chahiye, utility constant rakhte hue:

$$MRS_{1,2} = -\frac{dx_2}{dx_1}\bigg|_{u=\text{const}} = \frac{MU_1}{MU_2},$$

jahaan $MU_i = \partial u / \partial x_i$ good i ka marginal utility hai.

Derivation. $u(x_1, x_2) = \text{const}$ se, total differential:

$$\frac{\partial u}{\partial x_1} dx_1 + \frac{\partial u}{\partial x_2} dx_2 = 0 \Rightarrow \frac{dx_2}{dx_1} = -\frac{MU_1}{MU_2}.$$

MRTS vs. MRS — Twin Concepts

- **MRTS** (production) = MP_L / MP_K — isoquant ka slope.
- **MRS** (consumption) = MU_1 / MU_2 — indifference curve ka slope.

Mathematics identical hai; economic story dual hai (production vs. consumption).

4.8.1 Diminishing MRS

Preferences ki strict convexity $\Rightarrow$ indifference curve ke saath diminishing MRS. Intuition: jaise tum good 1 zyada consume karte ho, margin par use kam value dete ho (good 2 ke relative), aur use paane ke liye good 2 ki kam quantity chhodoge.

4.9 Canonical Utility Functions

Teen forms jo Weeks 2–3 ke teen production functions ko parallel karte hain:

4.9.1 (1) Perfect Substitutes (Linear)

$$u(x_1, x_2) = ax_1 + bx_2.$$

MRS = a/b constant; ICs parallel straight lines.

Common Mistake

“Perfect substitutes” ka matlab 1 : 1 exchange ratio *nahi* hai. Matlab MRS constant hai. Rs 100 dimes mein aur Rs 100 quarters mein perfect substitutes ho sakte hain, chahe ek quarter $\neq$ ek dime.

4.9.2 (2) Perfect Complements (Leontief)

$$u(x_1, x_2) = \min\{ax_1, bx_2\}.$$

ICs L-shaped (ray $bx_2 = ax_1$ ke saath right-angle kinks). Goods fixed proportions mein hi consume hote hain (left shoe + right shoe; do quarters + ek dime vending machine ke liye).

4.9.3 (3) Cobb–Douglas

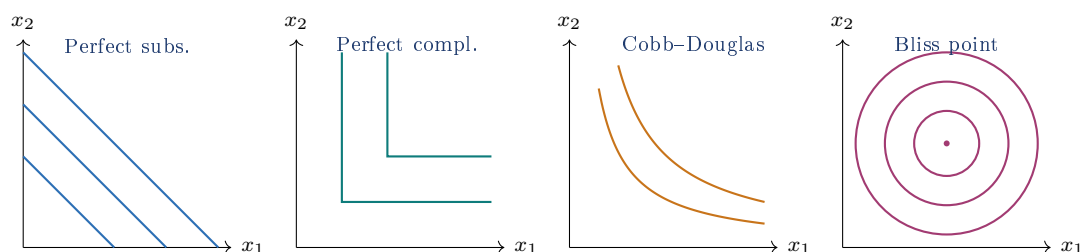
$$u(x_1, x_2) = x_1^\alpha x_2^\beta.$$

ICs smooth convex curves; MRS = $\frac{\alpha x_2}{\beta x_1}$ (diminishing).

4.9.4 (4) “Bliss Point” / Quadratic

$$u(x_1, x_2) = -(x_1 - a)^2 - (x_2 - b)^2.$$

ICs bliss point (a, b) ke around concentric circles. *More is not always better!* Yeh monotonicity violate karta hai par ek “satiation” preference illustrate karta hai.



PYQ Family — Coke-Machine (Leontief / Perfect Complement)

Story. Coke machine ko exactly 2 quarters + 1 dime per drink chahiye. Elmo ke paas 4 quarters aur 2 dimes hain.

Kitne drinks? # drinks = $\min\{q/2, d/1\} = \min\{4/2, 2/1\} = \min\{2, 2\} = 2$. **2 drinks.**

Kya Elmo ki preferences convex hain? Yes. Leontief preferences ke upper-contour sets weakly convex hote hain (L-shaped IC, par kink ke north-east ka region convex). *Common mistake:* “goods substitute nahi karte” ko “not convex” samajhna — non-substitutability convexity ke saath consistent hai.

PYQ Family — Diminishing Marginal Utility (2025T2 Q261–262), $U = y\sqrt{x}$

$$MU_x = \frac{\partial U}{\partial x} = \frac{y}{2\sqrt{x}}, \quad MU_y = \frac{\partial U}{\partial y} = \sqrt{x}. \text{ } \text{Dono} > 0 \text{ (more is better, monotonicity).}$$

Q261 (correct alternative): “more is better for each good.” (MU_x x par depend karta hai, par MU_y y par nahi — toh “MU of x independent of x ” wala part false hai.)

Q262 (incorrect alternative): $\frac{\partial MU_x}{\partial x} = -\frac{y}{4x^{3/2}} < 0$ (diminishing MU of x , *true*); $\frac{\partial MU_y}{\partial y} = 0$ (MU of y *does not* diminish). Toh “diminishing MU of y ” **incorrect** hai.

PYQ Family — Preference Axiom Violations (2024T1)

- “Cannot compare two goods” $\Rightarrow$ violates **Completeness** (completeness kehta hai koi bhi do bundles compare ho sakte hain).
 - Mango $\succ$ Apple, Apple $\succ$ Guava, Guava $\succ$ Mango $\Rightarrow$ violates **Transitivity** (cycle $M > A > G > M$; transitivity $M > A, A > G \Rightarrow M > G$ demand karti hai, par $G > M$ diya hai).
- Trap:** cyclical $\Rightarrow$ transitivity; “unable to compare” $\Rightarrow$ completeness. Mix mat karo.

4.10 How to Write This Unit in the Exam

Answer-Writing Strategy

- **Keywords:** budget line ($P_1x_1 + P_2x_2 = I$), slope $-P_1/P_2$, intercepts I/P_1 & I/P_2 , completeness, transitivity, reflexivity, continuity, monotonicity, convexity, ordinal utility, $MRS = MU_1/MU_2$, perfect substitutes/complements.
- **Diagram suggestions:** (i) budget line with intercepts + slope (Robinson Crusoe); (ii) canonical IC shapes (straight / L-shaped / convex / concentric circles); (iii) budget-line shift vs. rotation.
- **Axioms ko precisely state karo;** violation wale Qs mein *kaunsa* axiom toota hai woh naam lo (cycle $\Rightarrow$ transitivity).

Common Mistakes to Avoid

- “Perfect substitutes” ko 1 : 1 exchange samajhna — matlab MRS *constant*.
- Cardinal utility statements likhna (“twice as happy”) — utility *ordinal* hai.
- Leontief preferences ko “non-convex” kehna — woh weakly convex hote hain.
- Budget-line mein P_1 change par poora line parallel shift karna — woh *rotation* hai (intercept ek hi pivot karta hai).

4.11 Unit Summary — Quick Revision

Week 4 Quick-Revision Sheet

SR vs LR cost (continued):

- $SRTC \geq LRTC$, equality at long-run optimal $\bar{K}$ for that Q .
- $MC_{SR} = w/MP_L$. LRAC = lower envelope of SRAC curves.

Consumer Theory:

- Budget line: $P_1x_1 + P_2x_2 = I$; slope $-P_1/P_2$; intercepts $I/P_1, I/P_2$.
- Rationality (completeness, transitivity, reflexivity) + continuity $\Rightarrow$ utility function exists (Debreu).
- Utility ordinal: monotone transformations ke under invariant.
- Monotonicity $\Rightarrow$ ICs thin, downward-sloping, non-intersecting.
- Convexity $\Rightarrow$ ICs origin ki taraf jhuke; balanced bundles preferred.
- $MRS = MU_1/MU_2$.

Canonical Utility Functions: linear (perfect substitutes), Leontief (perfect complements), Cobb–Douglas, bliss-point (concentric circles).

Most Important Exam Topics

1. Budget line ka slope aur intercepts compute karna.
2. P_1, P_2, I change ka budget line par effect.
3. Description se perfect substitutes / complements identify karna.
4. Convex vs. non-convex preferences (Coke-machine: 2 quarters + 1 dime).
5. Utility function se MRS compute karna.

Frequently Asked PYQs

1. Coke-machine: $\min\{4/2, 2/1\} = 2$ drinks; preferences convex (yes).
2. Budget line (coconut-fish): slope -2 , intercepts $50/100$, halving effects.
3. $U = y\sqrt{x}$: more-is-better correct; diminishing MU of y incorrect.
4. Axiom violations: incomparable $\Rightarrow$ completeness; cycle $\Rightarrow$ transitivity.

Expected Long Questions

1. Budget line derive karo, intercepts/slope label karo, aur income vs. price changes ka effect (shift vs. rotation) diagram ke saath dikhao.
2. Rationality axioms + continuity state karke Debreu's theorem (utility existence) explain karo; ordinal utility ka matlab batao.
3. Chaar canonical utility functions, unki ICs aur MRS describe karo (diagrams ke saath), aur production functions ke saath duality dikhao.

Expected Short Questions

1. Budget line ka slope aur intercepts likho.
2. Monotonicity aur local non-satiation mein difference.
3. $u = xy$ ke liye MRS derive karo.
4. Perfect substitutes vs. perfect complements — ek-ek example.

One-Night-Before-Exam Revision Notes

- Budget: $P_1x_1 + P_2x_2 = I$; slope $-P_1/P_2$; intercepts $I/P_1, I/P_2$.
- Income $\uparrow$ parallel shift; $P_i \uparrow$ rotation (uss intercept pe pivot).
- Rational = complete + transitive + reflexive; + continuity $\Rightarrow$ utility.
- Utility ordinal; monotonic transform same preferences.
- $MRS = MU_1/MU_2$; perfect subs = constant MRS; Leontief = L-shaped, convex.

Quick Formula Sheet (Week 4)

$$\text{Budget: } P_1x_1 + P_2x_2 = I, \quad \text{slope} = -\frac{P_1}{P_2}, \quad \text{intercepts } \frac{I}{P_1}, \frac{I}{P_2}.$$

$$MU_i = \frac{\partial u}{\partial x_i}, \quad MRS_{1,2} = \frac{MU_1}{MU_2} = -\frac{dx_2}{dx_1} \Big|_{u=\bar{u}}, \quad \text{Optimum: } MRS_{1,2} = \frac{P_1}{P_2}.$$

$$\text{Perfect subs } u = ax_1 + bx_2, \quad \text{Leontief } u = \min\{ax_1, bx_2\}, \quad \text{Cobb-Douglas } u = x_1^\alpha x_2^\beta.$$

Practice Questions — Week 4

1. Budget line: $P_x = 5, P_y = 10, I = 100$. Draw karke intercepts aur slope label karo.
2. Agar P_x double ho jaaye, budget line ka kya hota hai?
3. $u(x, y) = xy$ ke liye $(2, 8)$ par MRS compute karo.
4. Kya dimes aur quarters Elmo ke liye convex preferences hain (jise per drink exactly 2 quarters + 1 dime chahiye)?
5. $u = 3x + 2y$ (perfect substitutes) ke liye MRS nikalo.